

Bounded-Support Hahn Arithmetic

Cofinal descent, optimal support thresholds,
maximal transcendence, and a Galois rank dichotomy

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Abstract

Let G be a nonzero set-sized ordered abelian group, let K have characteristic zero, and let $\mathcal{B}_G(K)$ be the ring of Hahn series whose support is bounded above in G . Put $\mathcal{F}_G(K) = \text{Frac } \mathcal{B}_G(K)$ and $\kappa = \text{cf}(G)$. We construct 2^κ elements of $K((t^G))$, all with positive integer coefficients and one common support of order type κ , that are algebraically independent over $\mathcal{F}_G(K)$. No smaller support cardinality can support even one transcendental element over this base. Every valuation ball contains such an independent family. In particular, the transcendence degree is the continuum for real-exponent Hahn fields over $\mathbb{R}$ or $\mathbb{C}$.

A complementary descent theorem proves that, whenever $H \subseteq G$ is cofinal and $K \subseteq L$, the fields $K((t^H))$ and $\mathcal{F}_G(L)$ are linearly disjoint over $\mathcal{F}_H(K)$. Polynomial relation ideals and finite algebraic degrees therefore survive simultaneous enlargement of coefficients and cofinal exponent groups. We also classify when $\mathcal{B}_G(K)$ is a field, reduce its unit theory to the highest Archimedean quotient, and prove a conditional GCD-descent theorem relevant to a published pre-Schreier question. Concrete Conway-normal-form specializations give independent families of actual surreal numbers, including maximal families of size 2^κ at every infinite regular cardinal.

A second part, from a second manuscript, develops the Galois theory of $\mathcal{F}_G(\mathbb{C})$ and $\mathcal{F}_G(\mathbb{R})$, which are also the fraction fields of the canonical omnific subrings at the scale G . For divisible G there is an exact rank dichotomy: without an order unit $\mathcal{F}_G(\mathbb{C})$ is algebraically closed and $\mathcal{F}_G(\mathbb{R})$ real closed; with one, every product of at most continuum many finite groups is a Galois group over $\mathcal{F}_G(\mathbb{C})$ inside the ambient Hahn field, with explicit independent radical towers and an explicit S_m family. Cofinal inclusions of exponent groups preserve every finite Galois extension, while noncofinal ones split every finite extension after scalar extension; an explicit countable tower inside $\mathbf{No}[i]$ repeats this.

Status. Complete arguments for the stated new results are given relative to explicitly identified standard Hahn-field and Conway-normal-form foundations. The proposed novelty is not a certified priority claim. This draft is not refereed or Lean-verified. The GCD application is explicitly conditional and is not used in any transcendence theorem.

In this collection. A two-source report: Sections 1–10 and the appendices come from the first manuscript, Sections 11–19 (the Galois part, added in batch 32) from the second; every label carries the prefix `bst:`, and those of the Galois part the sub-prefix `bst:gr:`. Its relation to the neighbouring reports is Sections 9 and 18.10; its provenance is Sections 1.4 and 11.1.

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1 The problem, the results, and their provenance

A Hahn series can contain infinitely many terms while its exponents remain in one bounded interval. Such a series is not a polynomial, and adjoining all such series produces a much larger base than a rational function field. The question pursued here is what remains genuinely transcendental after *every* bounded-support series has been admitted as a coefficient, and how that answer changes when the ambient exponent group changes.

Write

$$\mathcal{B}_G(K) = \{f \in K((t^G)) : \text{supp}(f) \text{ is bounded above in } G\}, \quad \mathcal{F}_G(K) = \text{Frac } \mathcal{B}_G(K). \quad (1)$$

The phrase “bounded support” always refers to the order on the specified exponent group. It does not mean finite support, bounded coefficients, or a bound on the cardinality of the support. In particular it is not the cardinal restriction sometimes called a λ -bounded Hahn field.

1.1 The principal theorems

Here is the theorem package in a form that can be read before its proofs. All fields and groups in these statements are sets. Cardinal assertions are made in set theory with choice.

Theorem 1.1 (Optimal-support independence; proved in Section 5). *Let K have characteristic zero and $G \neq 0$ be an ordered abelian group. Set $\kappa = \text{cf}(G)$. There are a strictly increasing cofinal sequence $(a_\alpha)_{\alpha < \kappa}$ of positive elements of G and positive integers $c_A(\alpha)$, for $A \subseteq \kappa$, such that*

$$Y_A = \sum_{\alpha < \kappa} c_A(\alpha) t^{a_\alpha} \quad (A \subseteq \kappa)$$

are algebraically independent over $\mathcal{F}_G(K)$. Their common support has order type κ . The minimum possible cardinality, and the minimum possible order type, of the support of a series transcendental over $\mathcal{F}_G(K)$ are both κ . Every nonempty valuation ball contains a family of 2^κ elements algebraically independent over $\mathcal{F}_G(K)$.

The strength is not a cardinality comparison between the base and the larger field: they can have the same cardinality. It is the construction of a free polynomial algebra over the entire bounded-support fraction field, with integer-coefficient witnesses and an optimal support requirement.

Theorem 1.2 (Cofinal coefficient–exponent descent; proved in Section 3). *Let $K \subseteq L$ be fields and let $0 \neq H \subseteq G$ be ordered abelian groups. If H is cofinal in G , multiplication induces an injection*

$$\mathcal{F}_G(L) \otimes_{\mathcal{F}_H(K)} K((t^H)) \longrightarrow L((t^G)).$$

Consequently

$$\mathcal{F}_G(L) \cap K((t^H)) = \mathcal{F}_H(K).$$

For every finite tuple from $K((t^H))$, its polynomial relation ideal over $\mathcal{F}_G(L)$ is the ideal generated by its relations over $\mathcal{F}_H(K)$. Finite algebraic degrees are preserved. In characteristic zero the cofinality hypothesis is also necessary for the displayed multiplication map to be injective.

For example, with $H = \mathbb{Z}$, $G = \mathbb{R}$, $K = \mathbb{Q}$, and $L = \mathbb{C}$, every algebraic relation among ordinary rational-coefficient Laurent series over $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$ already comes from relations over $\mathbb{Q}(t)$. This is a simultaneous coefficient and exponent extension statement, not merely an intersection statement.

The structural companion is an exact distinction between the existence and absence of a highest scale. The ring $\mathcal{B}_G(K)$ is a field precisely when G has no order unit. In that case it is the union of the Hahn fields over proper convex subgroups. For divisible G and real closed K , that union is real closed; for algebraically closed K of characteristic zero, it is algebraically closed. When an order unit exists, the units are controlled by the highest Archimedean quotient; see Section 7.

1.2 What is already known, and what is proposed here

L’Innocente and Mantova study bounded-support localization and its arithmetic in [1]. Their Proposition 2.4.5 already gives a cofinality obstruction to representing all Hahn series by fractions of one-sided series. That nonrationality obstruction is *not* claimed here as new. The proposed strengthening is the $2^{\text{cf}(G)}$ -sized algebraically independent family with one optimal common support, together with the coefficient–exponent descent and its exact converse. Their Proposition 3.5.1 supplies the upper-support multiplicativity used only in the unit and GCD discussion.

The repository audit used the documentation catalogue and the closest transcendence report at revision

048b72cf7cbfc8ab246e4f73788c10460cb3f6e0.

At that revision the catalogue listed 36 reports. Raw placement in commit 30dfb4f added this report and three others, bringing that historical count to 40; their editorial assembly followed in commit ed88b8f. Its tail-span report [5] constructs differential independence using independent square classes in coefficient extensions. Here the coefficients of the witnesses can all be positive integers, the base is a bounded-support fraction field, and the mechanism is support separation. These are different base fields and different assertions; neither report should be described as automatically subsuming the other. The catalogue also contains entire-function and cofinality results [6], but a ring of entire functions is not the ring in (1). Section 9 states precisely which results of these reports overlap with those here and which differ.

The proposed priority claims are deliberately limited. Targeted searches did not identify the combined descent theorem or the sharp independent-family theorem in these formulations. This is not an exhaustive MathSciNet, Zentralblatt, thesis, or citation-network audit. The elementary coding lemma, standard linear-disjointness algebra, Hahn field construction, real/algebraic closedness criteria, and Conway normal form are background, not new discoveries. A later repository revision, 6e34651fe640, was observed during the audit; its recorded change concerned Hahn–Hilbert spectral theory. No claim of a full source-by-source audit of that later revision is made.

1.3 Dependencies and the boundary of the claim

The main independence proof uses only Hahn multiplication, finite polynomial algebra, cardinal cofinality, and a fully proved finite-pattern coding lemma. It does not use a GCD theorem, the upper-support multiplicativity result, any surreal derivation, or analytic convergence. The descent proof is coefficientwise and works in arbitrary characteristic. Standard Hahn-field closedness and Conway normal form are invoked only where named. Appendix A proves an implication toward a published GCD/pre-Schreier problem; it does not supply the missing real-exponent GCD input.

1.4 Provenance of this report

Sections 1–10 and the appendices are built from one manuscript (source 01), *Bounded-Support Hahn Arithmetic: cofinal descent, optimal support thresholds, and maximal transcendence*, a research draft prepared with ChatGPT and dated 22 September 2026. It arrived as item 07 of the nineteenth batch of external manuscripts. Commit 30dfb4f placed the raw source and artifacts; commit ed88b8f assembled the article and added its first repository PDF. Its own repository audit is pinned to revision 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0 (Appendix B). That manuscript contributed every theorem, proof, example, appendix and non-claim printed in Sections 1–10 and Appendices A–B. Until batch 32 nothing was merged from another source. Sections 11–19 were added in batch 32 from a second manuscript (source 02) on the Galois theory of the same fraction fields; Section 11.1 gives

its provenance, the results of Sections 1–10 that it re-derives (each printed once and credited), and where the merge had to choose.

The editorial changes made when the report entered the collection are: every label received the prefix `bst :`; the actual surreal witnesses of Section 8, written ξ_A and ξ_p in the manuscript, are written η_A and η_p here, because the tail-span report already uses ξ_A for different numbers (Section 9.1); the terminology conventions of Section 2.3 and the comparison of Section 9 were added; and statements about the repository that had become stale were corrected, keeping the original pin as provenance. The delivered programs and records are kept byte-identical in code/ and data/; Section 10 gives the file names. No theorem statement was strengthened or weakened at assembly. The subsequent proof review expands the arguments and corrects the convergence and comparison wording. It also replaces the former 52c7ab6 placement citation, which is unavailable in the local history, by the distinct placement and assembly commits above. The editorial conventions of the Galois part (its notation, which renames several of source 02's symbols, and its [merge] marks) are stated in Sections 11.1 and 11.2.

2 Hahn series and bounded-support algebra

2.1 Conventions

An ordered abelian group is written additively. A Hahn series over K is a formal expression

$$f = \sum_{g \in G} f_g t^g$$

whose support $\{g : f_g \neq 0\}$ is well ordered. Addition is coefficientwise and multiplication is the Hahn convolution. At each exponent the relevant sum is finite. Finite unions and finite sumsets of well-ordered supports are well ordered. These facts give its ring operations. For a nonzero series, factor out its leading monomial to write $f = at^g(1 + h)$ with $a \in K^\times$ and h of strictly positive support (or $h = 0$). The Hahn–Neumann positive-support lemma makes $\sum_{n \geq 0} (-h)^n$ strongly summable and permits coefficientwise telescoping, giving the inverse $a^{-1}t^{-g} \sum_{n \geq 0} (-h)^n$. This standard inverse construction completes the field $K((t^G))$; see [1, Section 2.1]. It uses no valuation convergence of the finite geometric sums. We use the conventional valuation

$$v(f) = \min \text{supp}(f) \quad (f \neq 0), \quad v(0) = +\infty.$$

Thus $v(fg) = v(f) + v(g)$ and $v(f + g) \geq \min(v(f), v(g))$. The convention at zero is stated explicitly because some treatments of one-sided series use a different special-symbol convention.

A family of Hahn series is *strongly summable* when its combined support is well ordered and only finitely many summands contribute at any exponent. A sum along a strictly increasing sequence of exponents is strongly summable, independently of whether its finite partial sums converge in a chosen topology. All transfinite sums below are Hahn sums in this sense.

For $f \in K((t^G))$ and $\delta \in G$, write

$$f_{\leq \delta} = \sum_{g \leq \delta} f_g t^g.$$

A nonzero Hahn support has a least element. Therefore “bounded above” in (1) is equivalent to containment in some order interval $[\ell, u] \subseteq G$.

Lemma 2.1 (Elementary support bounds). *The set $\mathcal{B}_G(K)$ is a subring of $K((t^G))$, contains K and every monomial, and is closed under taking arbitrary subseries. If $\text{supp}(f) \subseteq [\ell, u]$ and $\text{supp}(g) \subseteq [\ell', u']$, then*

$$\text{supp}(fg) \subseteq [\ell + \ell', u + u'].$$

Every finite family of bounded-support series has common lower and upper support bounds.

Proof. Every exponent of a product is a sum of one exponent from each factor. Cancellation only removes terms and cannot create an exponent outside the indicated interval. A finite union is bounded by the minimum of its lower bounds and the maximum of its upper bounds. Negation leaves the support unchanged. Every subset of a well-ordered set is well ordered, proving the subseries assertion. These observations also cover zero by using arbitrary bounds for its empty support. $\square$

Proposition 2.2 (Localization and density). *For any nonzero G ,*

$$\mathcal{B}_G(K) = t^G K((t^{G^{\leq 0}})).$$

Both $\mathcal{B}_G(K)$ and $\mathcal{F}_G(K)$ are dense in $K((t^G))$ for the valuation topology.

Proof. If $\text{supp}(f) \leq u$, then $t^{-u}f$ has nonpositive support. Conversely a monomial translate of a nonpositive support is bounded above. For density, $f_{\leq \delta} \in \mathcal{B}_G(K)$ and either $f = f_{\leq \delta}$ or $v(f - f_{\leq \delta}) > \delta$. Hence every basic valuation ball contains a bounded-support element. The assertion for the fraction field follows. Here $t^G K((t^{G^{\leq 0}}))$ denotes the set of monomial translates of the nonpositive-support subring. It is its localization at the monomials t^{-g} for $g \geq 0$: choose the upper bound $u \geq 0$ above, so $f = (t^{-u}f)/t^{-u}$. This specifies the localization used in Appendix A. $\square$

There is no assertion here that bounded-support series are finite sums. For example

$$\sum_{n \geq 2} t^{1-1/n} \in \mathcal{B}_{\mathbb{Q}}(\mathbb{R})$$

has infinitely many exponents in $[1/2, 1)$. Nor is density an algebraicity statement. The main theorem will show that an enormous transcendental extension can remain after adjoining this dense subring's entire fraction field.

2.2 Cofinality

Definition 2.3. For a nonzero ordered abelian group G , its cofinality $\text{cf}(G)$ is the least cardinality of a subset unbounded above in G . An ordered subgroup H is *cofinal* if for every $g \in G$ some $h \in H$ satisfies $h > g$. An *order unit* is an element $u > 0$ such that for every $g \in G$ some positive integer n satisfies $|g| \leq nu$.

Lemma 2.4. *The cardinal $\kappa = \text{cf}(G)$ is infinite and regular. There is a strictly increasing positive cofinal sequence $(g_\alpha)_{\alpha < \kappa}$. Every subset of G of cardinality less than κ is bounded above. An ordered subgroup is either cofinal or bounded above.*

Proof. No finite subset is cofinal, since it has a maximum and G has positive elements. The last two assertions follow from the definitions. Start with a cofinal family of size κ and recursively choose a value exceeding its next member and all previously chosen values. At stage $\alpha < \kappa$ the latter set has size less than κ , so a bound exists. This gives the increasing sequence. If κ were singular, a cofinal subsequence of its indices of cardinality $\text{cf}(\kappa) < \kappa$ would give a smaller cofinal subset of G , a contradiction. $\square$

For a nonzero subgroup of $\mathbb{R}$, every positive element is an order unit and the cofinality is $\aleph_0$. The converse implication from countable cofinality to the existence of an order unit is false: a countable ascending union of strictly increasing Archimedean scales has no highest scale. Section 8 gives concrete examples inside **No**.

2.3 Terminology shared with other reports

The following conventions fix one meaning for each word and symbol that is used differently elsewhere in the collection.

1. *Cofinality*. $\text{cf}(G)$ is the least cardinality of a subset of the ordered set G that is unbounded above. This is exactly the invariant $\text{cf}(\Gamma)$ of the entire-functions report [6], whose definition uses “every element is at most some member”; for a nonzero ordered group, which has no greatest element, the two definitions agree.
2. *Cofinal*. The adjective is used for a subset or sequence of G that is unbounded above, and for an ordered subgroup $H \subseteq G$ with the same property. It is used once for a set of *indices*: in Lemma 5.1 a set of ordinals is cofinal in the initial ordinal κ . The entire-functions report expressly excludes that ordinal-index meaning from its own word (its remark `ent : rem : one-name`); here it is confined to that lemma and the proofs that cite it.
3. *Order unit*. The definition above agrees with `ent : def : order-unit`: $|g| \leq nu$ for all g is equivalent to $\{\mu : m \geq 1\}$ being unbounded above. An order unit forces $\text{cf}(G) = \aleph_0$, and the converse fails, in both reports.
4. *Cofinite* is not *cofinal*. The tail-span report’s cofinite span is an intersection over finite deletions of coefficient indices; nothing in this report deletes finitely many indices.
5. *Bounded*. “Bounded support” always means bounded above in the named exponent group, as in (1), and never bounded above in all of **No** (Section 8). It is not a bound on coefficients or on valuations, and not the cardinal restriction of a λ -bounded Hahn field.
6. *Symbols*. $\mathcal{B}_G(K)$ and $\mathcal{F}_G(K)$ always carry both the exponent group and the coefficient field. The collection’s notation guide [8] writes F_Γ for the full real Hahn field $\mathbb{R}((t^\Gamma))$, and the tail-span report writes B and B_0 for full Hahn fields $M((t^\Gamma))$ and $\mathbb{Q}((t^\mathbb{Q}))$. Neither is meant by the calligraphic letters here: $\mathcal{F}_G(K)$ is the fraction field of the bounded-support ring, a proper subfield of $K((t^G))$ in characteristic zero.

3 Cofinal coefficient–exponent descent

The following result is useful independently of the lacunary constructions. It explains why a bounded-support coefficient cannot conceal a new relation among series supported on a cofinal subgroup.

3.1 Coset projections and coefficient functionals

Let $K \subseteq L$ be fields, $H \subseteq G$ an ordered subgroup, and $r \in G$. Define the shifted coset projection

$$\pi_r(f) = \sum_{h \in H} f_{r+h} t^h \in L((t^H)). \quad (2)$$

It is defined for every $f \in L((t^G))$: a subset of its support remains well ordered, and translation preserves that property. This is not asserted to be a ring homomorphism: for $H = \mathbb{Z} \subset G = \mathbb{Q}$, one has $\pi_0(t^{1/2}) = 0$ but $\pi_0((t^{1/2})^2) = t$. Its useful property is the module identity below.

Lemma 3.1 (Projection identity). *For $z \in L((t^H))$ and $b \in L((t^G))$,*

$$\pi_r(bz) = \pi_r(b)z.$$

If H is cofinal and $b \in \mathcal{B}_G(L)$, then $\pi_r(b) \in \mathcal{B}_H(L)$.

Proof. A term of bz at an exponent in $r + H$ can only use a term of b in that same coset, because all exponents of z lie in H . At every exponent the convolution is finite, so selecting the coset commutes with it. If the support of b is bounded above by $u \in G$, then the shifted support is bounded above in G by $u - r$. Cofinality supplies an H -valued bound. $\square$

Let $\lambda : L \rightarrow K$ be any K -linear functional. Apply it to each coefficient to obtain

$$\lambda_* : L((t^H)) \longrightarrow K((t^H)).$$

This map can kill support terms but introduces none. In particular it preserves boundedness of support. For $z \in K((t^H))$ it satisfies

$$\lambda_*(bz) = \lambda_*(b)z. \quad (3)$$

The reason is again the finiteness of each convolution coefficient, now combined with K -linearity. No multiplicativity of λ is needed. For any nonzero $a \in L$, a K -linear functional with $\lambda(a) \neq 0$ exists: extend a to a basis and prescribe its value to be 1.

3.2 The descent theorem

Theorem 3.2 (Simultaneous descent). *Let $K \subseteq L$ be fields and let $0 \neq H \subseteq G$ be cofinal. Every finite family of elements of $K((t^H))$ linearly independent over $\mathcal{F}_H(K)$ remains linearly independent over $\mathcal{F}_G(L)$. Equivalently, the natural multiplication map*

$$\mathcal{F}_G(L) \otimes_{\mathcal{F}_H(K)} K((t^H)) \longrightarrow L((t^G)) \quad (4)$$

is injective.

Proof. Let $z_1, \dots, z_m \in K((t^H))$ be linearly independent over $\mathcal{F}_H(K)$. Suppose there is a nonzero linear relation over $\mathcal{F}_G(L)$. After multiplying by a common nonzero denominator, it becomes

$$\sum_{j=1}^m b_j z_j = 0, \quad b_j \in \mathcal{B}_G(L), \quad (5)$$

with some $b_{j_0} \neq 0$. Choose an exponent $r \in \text{supp}(b_{j_0})$ and set $a = (b_{j_0})_r \neq 0$. Applying π_r to (5) gives

$$\sum_j \pi_r(b_j) z_j = 0,$$

where all projected coefficients belong to $\mathcal{B}_H(L)$. Choose a K -linear functional $\lambda : L \rightarrow K$ with $\lambda(a) = 1$. Applying λ_* and using (3) yields

$$\sum_j \lambda_*(\pi_r(b_j)) z_j = 0.$$

Its coefficients belong to $\mathcal{B}_H(K) \subseteq \mathcal{F}_H(K)$. The coefficient of t^0 in $\lambda_*(\pi_r(b_{j_0}))$ is 1, so the relation is nonzero. This contradicts the assumed independence.

To verify the tensor formulation directly, write a tensor using a finite $\mathcal{F}_H(K)$ -linearly independent family of second factors. If its image is zero, the first part of the proof forces every first-factor coefficient to vanish. The converse equivalence follows by the same representation. This agrees with the standard definition of linear disjointness [3]. $\square$

Corollary 3.3 (Intersection, relations, and algebraic degree). *Under the hypotheses of Theorem 3.2:*

1. $\mathcal{F}_G(L) \cap K((t^H)) = \mathcal{F}_H(K)$ inside $L((t^G))$.
2. For a finite tuple z in $K((t^H))$, let $I_K(z)$ and $I_L(z)$ be its relation ideals over $\mathcal{F}_H(K)$ and $\mathcal{F}_G(L)$, respectively. Then

$$I_L(z) = I_K(z) \mathcal{F}_G(L)[X_1, \dots, X_m].$$

3. Algebraic independence of any family in $K((t^H))$ is preserved in both directions by this base enlargement.
4. If $z \in K((t^H))$ is algebraic of degree d over $\mathcal{F}_H(K)$, it has degree d over $\mathcal{F}_G(L)$, with the same monic minimal polynomial.

Proof. For the intersection statement, $z \notin \mathcal{F}_H(K)$ makes $1, z$ linearly independent over $\mathcal{F}_H(K)$, hence over $\mathcal{F}_G(L)$; it cannot then belong to $\mathcal{F}_G(L)$. The reverse inclusion is immediate.

Let $E = \mathcal{F}_H(K)$ and $F = \mathcal{F}_G(L)$. The quotient $E[X]/I_K(z)$ embeds in $K((t^H))$. Tensoring that inclusion with the E -vector space F preserves injectivity. By Theorem 3.2, the resulting algebra still embeds in $L((t^G))$. Its presentation is $F[X]/I_K(z)F[X]$, so its evaluation kernel is exactly the asserted ideal. This proves the second assertion and then the third. Finally $1, z, \dots, z^{d-1}$ remain linearly independent, while the old degree- d equation still holds, proving the last assertion. $\square$

For a finite Galois subextension of $K((t^H))/\mathcal{F}_H(K)$, the same argument preserves the degree and Galois group after compositum with $\mathcal{F}_G(L)$. Indeed, the tensor product embeds in the Hahn field by descent, so it is a domain. It is finite-dimensional over $\mathcal{F}_G(L)$ of the original degree, hence a field: multiplication by a nonzero element is an injective linear map and therefore surjective. Its distinct base-field automorphisms extend by the identity on $\mathcal{F}_G(L)$, and their number already equals that degree. This observation does not assert that every algebraic germ extension is Galois. (Added in batch 32.) Corollary 12.6 adds that the whole subgroup–intermediate-field correspondence is transported, and Sections 13–15 use this transport to realize every finite group, and products of at most c of them, as Galois groups over $\mathcal{F}_G(\mathbb{C})$ when G has an order unit.

3.3 The ordinary Laurent-series case

Corollary 3.4 (One cofinal scale). *If $u > 0$ is an order unit of G and $K \subseteq L$, then $K((t^u))$ and $\mathcal{F}_G(L)$ are linearly disjoint over $K(t^u)$. In particular*

$$K((t^u)) \cap \mathcal{F}_G(L) = K(t^u).$$

Proof. The subgroup $\mathbb{Z}u$ is cofinal. Its bounded well-ordered subsets are finite, so $\mathcal{B}_{\mathbb{Z}u}(K) = K[t^u, t^{-u}]$ and its fraction field is $K(t^u)$. Apply Theorem 3.2. $\square$

Example 3.5 (An algebraic germ does not become a bounded fraction). Take $s = t^u$ and characteristic zero. The binomial series

$$z = (1 + s)^{1/2} = 1 + \frac{1}{2}s - \frac{1}{8}s^2 + \frac{1}{16}s^3 - \dots \in \mathbb{Q}[[s]]$$

has degree two over $\mathcal{F}_G(L)$ for every characteristic-zero extension $\mathbb{Q} \subseteq L$ and every G in which u is an order unit. To check its degree over $\mathbb{Q}(s)$, observe that $1 + s$ is not a rational square: in a coprime quotient of polynomials, the multiplicity of the factor $s + 1$ would be even on a square and odd on $1 + s$. Corollary 3.3 then preserves degree two. Thus it is algebraic but not itself in the bounded-support fraction field. In contrast, $(1 - s)^{-1}$ is in that fraction field, although its support is unbounded.

(Added in batch 32.) In the notation of Section 14, $z = r_{-1,2}$. Proposition 14.2 and Theorem 14.3 extend the example to all roots $r_{a,n}$ of $1 - as$ and to their joint degrees, and Theorem 14.7 uses the case $a = 1, n = 2$ to show that the closedness of Corollary 7.2 fails whenever G has an order unit.

3.4 Why noncofinal enlargement behaves oppositely

If H is not cofinal, the entire group H is bounded above in G . Consequently

$$K((t^H)) \subseteq \mathcal{B}_G(L) \subseteq \mathcal{F}_G(L). \quad (6)$$

Every old Hahn series becomes a bounded-support coefficient in the enlarged workspace. Once Theorem 5.3 has established $\mathcal{F}_H(K) \neq K((t^H))$ in characteristic zero, this proves the converse in Theorem 1.2: choose z in that difference. The pair $1, z$ is independent over the smaller base but dependent over the larger one. Explicitly, $1 \otimes z - z \otimes 1$ is a nonzero tensor sent to zero by multiplication. To see that it is nonzero, extend $1, z$ to a basis of $K((t^H))$ over $\mathcal{F}_H(K)$ and use the linear functional taking 1 to zero and z to one; applying it to the second tensor factor gives 1 . The exceptional subgroup $H = 0$ is excluded because its Hahn field and fraction field are both just K .

This dichotomy is about the actual embedding of exponent groups. Merely saying that both groups are divisible or both have countable cofinality does not imply that the embedding is cofinal. (Added in batch 32.) Theorem 16.3 states the Galois form of this dichotomy for $K = L = \mathbb{C}$ and divisible H : a cofinal inclusion keeps every finite extension of $\mathcal{F}_H(\mathbb{C})$ a field of the same degree, and a noncofinal one splits every finite extension completely.

4 The support-separation mechanism

This section supplies the estimate that upgrades the earlier nonrationality obstruction to algebraic independence. It is entirely algebraic. Even when a truncation has infinitely many terms, its support has the bound used in the estimate.

4.1 Separated exponent systems

Definition 4.1. Let κ be an infinite cardinal. A *separated exponent system* in G consists of positive elements a_α, b_α for $\alpha < \kappa$ such that:

1. a_α is strictly increasing and cofinal, and $a_\beta \leq b_\alpha$ whenever $\beta < \alpha$;
2. for every positive integer d and every $w \in G$, eventually in α one has

$$a_\alpha > db_\alpha + w, \quad a_{\alpha+1} > da_\alpha + w. \quad (7)$$

Every infinite cardinal is a limit ordinal, so $\alpha + 1 < \kappa$ for all $\alpha < \kappa$.

Lemma 4.2 (Existence at the exact cofinality). *Every nonzero ordered abelian group admits a separated exponent system of length $\kappa = \text{cf}(G)$.*

Proof. Fix a strictly increasing positive cofinal sequence g_α from Lemma 2.4.

First suppose $\kappa = \aleph_0$, and index by $n \geq 0$. Choose $b_n > 0$ strictly exceeding g_n and, when $n > 0$, a_{n-1} . Set $a_n = (n+2)b_n$. These elements are increasing and cofinal, and earlier a_j are at most b_n . Given d, w , eventually $b_n > w$ and $n+2 \geq d+1$, so $a_n - db_n \geq b_n > w$. At the next index the same estimate gives $a_{n+1} > db_{n+1} + w \geq da_n + w$. Both inequalities hold eventually.

Now suppose κ is uncountable. At stage α , the earlier a_β , together with g_α , form a set of size less than κ . Choose a positive strict upper bound b_α for them. The countable set $\{nb_\alpha : n \geq 1\}$ also has size less than κ , so choose a_α strictly above every one of its elements. This gives a strictly increasing cofinal sequence. For fixed d, w , eventually $b_\alpha > w$; then $a_\alpha > (d+1)b_\alpha > db_\alpha + w$. The estimate at $\alpha+1$, and $b_{\alpha+1} > a_\alpha$, give the second inequality of (7). $\square$

Example 4.3 (Factorial spacing). If u is an order unit, the concrete choices

$$a_n = n!u, \quad b_n = (n-1)!u \quad (n \geq 2)$$

form a separated exponent system. For fixed d , $a_n - db_n = (n-d)(n-1)!u$ eventually exceeds every $w \in G$. The next-step inequality is checked in the same way.

4.2 A polynomial cannot bridge two separated support bands

Fix a separated exponent system $(a_\alpha, b_\alpha)_{\alpha < \kappa}$. Let

$$P(X) = \sum_v p_v X^v \in \mathcal{B}_G(K)[X_1, \dots, X_m]$$

be nonzero of total degree $d \geq 1$. Its top homogeneous part is denoted P_d . Choose $\ell, u \in G$ such that every p_v is supported in $[\ell, u]$. For coefficient vectors $c(\alpha) = (c_1(\alpha), \dots, c_m(\alpha)) \in K^m$, put

$$Y_j = \sum_{\alpha < \kappa} c_j(\alpha) t^{a_\alpha}.$$

Zero entries cause no problem: each support is a subset of the well-ordered set of a_α .

Lemma 4.4 (Support-band detection). *Suppose an index α satisfies*

$$a_\alpha > db_\alpha + (u - \ell), \quad a_{\alpha+1} > da_\alpha + (u - \ell), \quad (8)$$

and $P_d(c(\alpha)) \neq 0$ in $K((t^G))$. Then $P(Y_1, \dots, Y_m) \neq 0$.

Proof. Define the earlier part and the truncation through α by

$$A_j = \sum_{\beta < \alpha} c_j(\beta) t^{a_\beta}, \quad T_j = A_j + c_j(\alpha) t^{a_\alpha}.$$

Every A_j is supported in $[0, b_\alpha]$ and every T_j in $[0, a_\alpha]$. Expanding in an auxiliary *polynomial* variable Z , write

$$P(A_1 + c_1(\alpha)Z, \dots, A_m + c_m(\alpha)Z) = \sum_{k=0}^d Q_k Z^k. \quad (9)$$

The coefficient of Z^d is exactly $P_d(c(\alpha))$, so $Q_d \neq 0$. In a monomial of degree $|v|$, a contribution to Q_k uses k copies of Z and $|v| - k$ factors from the A_j . Reaching degree d in Z therefore uses only

top-degree monomials and no A_j factors. Every coefficient Q_k is a finite sum of products, each consisting of a p_v , a scalar, and at most d factors chosen from the A_j . Lemma 2.1 therefore gives

$$\text{supp}(Q_k) \subseteq [\ell, u + db_\alpha] \quad (0 \leq k \leq d).$$

After substituting $Z = t^{a_\alpha}$, these supports lie in the bands

$$[\ell + ka_\alpha, u + db_\alpha + ka_\alpha]. \quad (10)$$

The first inequality in (8) says that successive bands are strictly disjoint. Since $Q_d \neq 0$, choose $g \in \text{supp } Q_d$. The nonzero coefficient at $g + da_\alpha$ lies in the last band and receives no contribution from any earlier band. Hence $P(T_1, \dots, T_m) \neq 0$.

A direct bound using the original polynomial and the supports of T_j shows that

$$\text{supp}(P(T)) \subseteq [\ell, u + da_\alpha]. \quad (11)$$

Meanwhile every tail $Y_j - T_j$ is either zero or has valuation at least $a_{\alpha+1}$. Expand each difference $Y^\nu - T^\nu$ as a sum of products containing at least one such tail. All other series factors have nonnegative valuation. Thus

$$v(P(Y) - P(T)) \geq \ell + a_{\alpha+1}. \quad (12)$$

By the second inequality in (8), this lower bound is strictly above the upper bound in (11). The nonzero bounded-support series $P(T)$ cannot be canceled by the tail. Hence $P(Y) \neq 0$. $\square$

Remark 4.5 (What the proof does not assume). Neither $P(T)$ nor the earlier parts A_j need have finite support. No supremum of their supports in G is required; any displayed upper bound suffices. There is no rearrangement of a non-summable family and no appeal to a numerical limit. All polynomial expansions are finite, and each coefficientwise product is a Hahn convolution.

Corollary 4.6 (A one-series version). *Let (a_n) be positive and strictly increasing, and suppose that for every $d \geq 1$ and $w \in G$, eventually $a_{n+1} > da_n + w$. If every $c_n \in K$ is nonzero, then $Y = \sum_n c_n t^{a_n}$ is transcendental over $\mathcal{F}_G(K)$.*

Proof. For a putative relation clear denominators to obtain a nonzero polynomial $P \in \mathcal{B}_G(K)[X]$. The case of constant P is impossible. The same tail estimate as (12) implies $P(T_n) = 0$ for every sufficiently large n , since otherwise its bounded support could not be canceled. Here T_n is the finite truncation through n , supported at most at a_n . These truncations are pairwise distinct: the difference of two of them has the nonzero coefficient c_j at the first newly included exponent. A nonzero polynomial over the field $K((t^G))$ has only finitely many roots. This contradiction proves the claim. $\square$

The following comparison was added in batch 31. The independent-copies report [16] prints the same tail estimate and root count at every cofinality (its `isc:thm:cofinalgap`) and claims no priority for it. Let K be an arbitrary field, finite fields included, let $G \neq 0$ be set-sized, and let $(a_\alpha)_{\alpha < \text{cf}(G)}$ be a strictly increasing cofinal family in $G^{>0}$ such that, for every $d \geq 1$ and $w \in G$, eventually $a_{\alpha+1} > da_\alpha + w$. Such a family exists (its `isc:lem:gapscale`). Then $1 + \sum_{\alpha < \text{cf}(G)} t^{a_\alpha}$, with all coefficients 1, is transcendental over $\mathcal{F}_G(K)$. Corollary 4.6 also works over any field and allows arbitrary nonzero coefficients, but it is indexed by $\mathbb{N}$ and its hypothesis makes (a_n) cofinal, so it needs $\text{cf}(G) = \aleph_0$. Theorem 5.3 covers every cofinality but needs characteristic zero. The single-series statement at uncountable cofinality over a field of positive characteristic, in particular a finite field, is therefore not stated in this report. It does not touch the multivariable problem, and finite coefficient fields remain outside the independence theorem (Section 10).

The multivariable argument needs more than this root-count observation. One must ensure that the top homogeneous polynomial does not vanish on all chosen coefficient vectors. The next section supplies exactly that certificate, without any algebraic independence assumption on the coefficients themselves.

5 Independent coefficient codes and the sharp theorem

5.1 A finite-pattern coding lemma

The following elementary construction is an independent-family coding argument. Its role is to realize any prescribed finite tuple of positive integers at cofinally many positions. It is not being claimed as a new set-theoretic principle.

Lemma 5.1 (Cofinal finite-pattern realization). *For any infinite cardinal κ , there are functions*

$$c_A : \kappa \longrightarrow \mathbb{N}_{>0} \quad (A \subseteq \kappa)$$

such that for every finite list of distinct subsets $A_1, \dots, A_m$ and every $(q_1, \dots, q_m) \in \mathbb{N}_{>0}^m$, the set

$$\{\alpha < \kappa : c_{A_j}(\alpha) = q_j \text{ for all } j\}$$

has cardinality κ , and in particular is cofinal in κ .

Proof. Let D_κ consist of triples (s, h, ρ) where s is a finite subset of κ , $h : \mathcal{P}(s) \rightarrow \mathbb{N}_{>0}$ is a table, and $\rho < \kappa$. There are κ finite subsets of κ , countably many tables for each finite s , and κ choices for ρ . Hence $|D_\kappa| = \kappa$. Fix a bijection

$$\alpha \longmapsto (s_\alpha, h_\alpha, \rho_\alpha) \quad \text{from } \kappa \text{ to } D_\kappa$$

and set

$$c_A(\alpha) = h_\alpha(A \cap s_\alpha). \tag{13}$$

For each pair $A_j \neq A_k$, choose a point in their symmetric difference. The finite set s of these points makes $A_j \cap s$ pairwise distinct. A table h can therefore take the prescribed value q_j at each of them; set its remaining values to 1. Every triple (s, h, ρ) realizes the target tuple, independently of ρ . There are κ such triples, which proves the cardinality assertion. A bounded subset of the initial ordinal κ has cardinality smaller than κ , so the realizing indices are cofinal. Regularity of κ is not needed for this coding lemma; cardinality κ already rules out a bounded subset of its initial ordinal. For a single prescribed set, the separating set s may be empty. Distinct sets give distinct coefficient streams: prescribe values 1 and 2 at one common realizing index. $\square$

Lemma 5.2 (Integer grids detect polynomials). *Let E be a field of characteristic zero. A nonzero polynomial $Q \in E[X_1, \dots, X_m]$ is nonzero at some point of $\mathbb{N}_{>0}^m$. More precisely, if each variable has degree at most d , any product of m sets of $d + 1$ distinct elements of E contains a nonzero value.*

Proof. For no variables, the assertion concerns a nonzero constant and is immediate. For one variable, a nonzero polynomial of degree at most d has at most d roots. For the induction step, regard Q as a polynomial in its last variable. If it vanishes at every point of the product grid, then for each fixed tuple of the first $m - 1$ variables all its coefficients vanish, by the one-variable statement. The induction hypothesis applied to those coefficient polynomials makes each identically zero, a contradiction. The positive integers are distinct in a characteristic-zero field. $\square$

5.2 Independence over all bounded-support fractions

Theorem 5.3 (The sharp support theorem). *Let K have characteristic zero and let $0 \neq G$ be an ordered abelian group. Put $\kappa = \text{cf}(G)$. Choose a separated exponent system as in Lemma 4.2 and a coefficient code as in Lemma 5.1. Then*

$$Y_A = \sum_{\alpha < \kappa} c_A(\alpha) t^{a_\alpha} \quad (A \subseteq \kappa) \quad (14)$$

are algebraically independent over $\mathcal{F}_G(K)$. Consequently

$$\text{trdeg}_{\mathcal{F}_G(K)} K((t^G)) \geq 2^\kappa. \quad (15)$$

Every member has positive integer coefficients and exactly the common support $\{a_\alpha : \alpha < \kappa\}$, of order type κ .

Proof. First, each Y_A is a valid Hahn series: the strictly increasing map $\alpha \mapsto a_\alpha$ identifies its image with the well-ordered initial ordinal κ , and characteristic zero makes every prescribed positive integer coefficient nonzero. Thus its support is exactly that image. Fix distinct $A_1, \dots, A_m$ and a nonzero polynomial over $\mathcal{F}_G(K)$. Clear a common nonzero denominator to obtain a nonzero polynomial $P \in \mathcal{B}_G(K)[X_1, \dots, X_m]$. If its total degree is zero, its evaluation is already nonzero. Otherwise let its degree be $d \geq 1$ and choose common coefficient support bounds $[\ell, u]$.

Its top homogeneous part P_d is a nonzero polynomial over the field $K((t^G))$. By Lemma 5.2, there is a tuple of positive integers $q = (q_1, \dots, q_m)$ such that $P_d(q) \neq 0$. Lemma 5.1 realizes this exact tuple at cofinally many indices α . The gap inequalities (8) hold at every sufficiently large index. Choose a realizing index after that threshold. Lemma 4.4 then gives

$$P(Y_{A_1}, \dots, Y_{A_m}) \neq 0.$$

This handles every finite subfamily and every nonzero polynomial, which is algebraic independence. There are 2^κ subsets A , proving the lower bound. Finally the coefficients are positive integers, hence nonzero in K , so none of the designated exponents disappears. $\square$

For the remaining corollaries in this section, retain the hypotheses $\text{char } K = 0$, $G \neq 0$ and $\kappa = \text{cf}(G)$, and the family Y_A of Theorem 5.3.

Corollary 5.4 (Optimal cardinality and order type). *Among all $z \in K((t^G))$ transcendental over $\mathcal{F}_G(K)$, the least support cardinality is $\text{cf}(G)$ and the least support order type is the initial ordinal $\text{cf}(G)$.*

Proof. A support of cardinality less than $\kappa = \text{cf}(G)$ is bounded above by definition of cofinality, so its series already belongs to $\mathcal{B}_G(K)$. Theorem 5.3 supplies a transcendental series with support of cardinality and order type exactly κ . An ordinal smaller than the initial ordinal κ has smaller cardinality, proving the order-type assertion as well. $\square$

Thus countable supports suffice exactly when the value group has countable cofinality. When the cofinality is uncountable, every countable-support series is already a coefficient in the base, regardless of how sparse its support looks inside a smaller subgroup.

Corollary 5.5 (Arbitrary bounded perturbations). *For any choices $b_A \in \mathcal{F}_G(K)$ and $u_A \in \mathcal{F}_G(K)^\times$, the family $(u_A Y_A + b_A)_{A \subseteq \kappa}$ is algebraically independent over $\mathcal{F}_G(K)$. In particular arbitrary bounded-support changes, or deletion of an arbitrary bounded initial part of each series, preserve independence.*

Proof. On each finite subfamily the substitutions $X_A \mapsto u_A X_A + b_A$ define an invertible polynomial change of variables over the base field. They therefore send nonzero polynomials to nonzero polynomials. A deleted bounded initial part belongs to $\mathcal{B}_G(K)$. $\square$

Corollary 5.6 (Independent families in every valuation ball). *Every nonempty ball*

$$U(f, \delta) = \{z \in K((t^G)) : v(z - f) > \delta\}$$

contains 2^κ elements algebraically independent over $\mathcal{F}_G(K)$. It also contains an element of $\mathcal{B}_G(K)$.

Proof. Put $b = f_{\leq \delta} \in \mathcal{B}_G(K)$. Choose $\beta < \kappa$ with $a_\beta > \delta$, and define

$$Z_A = b + \sum_{\alpha \geq \beta} c_A(\alpha) t^{a_\alpha}.$$

Then $v(Z_A - b) > \delta$ and $v(b - f) > \delta$, with the usual convention for zero, so every Z_A lies in the ball. Each Z_A differs from Y_A by a bounded-support series: the omitted initial part is bounded above by a_β , and b is bounded. Corollary 5.5 proves their independence. The element b itself belongs to the same ball. $\square$

Remark 5.7 (A precise limitation of bounded truncation data). No single bounded truncation can determine algebraicity over $\mathcal{F}_G(K)$. The same truncation is shared by a base-field element and by every member of a family as in Corollary 5.6. This is an information-theoretic statement about those data, not a claim that every representation of Hahn series has an undecidable algebraicity problem.

5.3 Exact transcendence degrees

Corollary 5.8 (Cardinality-matched groups). *Suppose $\kappa = \text{cf}(G) = |G|$ and $|K| \leq 2^\kappa$, with K of characteristic zero. Then*

$$\text{trdeg}_{\mathcal{F}_G(K)} K((t^G)) = 2^\kappa.$$

Proof. The lower bound is Theorem 5.3. A Hahn series is a function from G to K , so

$$|K((t^G))| \leq |K|^{|G|} \leq (2^\kappa)^\kappa = 2^\kappa.$$

The cardinality of a field bounds its transcendence degree, proving the upper bound independently of the construction. $\square$

Corollary 5.9 (Every nonzero real-exponent subgroup). *Let $0 \neq G \subseteq \mathbb{R}$ and let K have characteristic zero and $|K| \leq \mathfrak{c} = 2^{\aleph_0}$. Then*

$$\text{trdeg}_{\mathcal{F}_G(K)} K((t^G)) = \mathfrak{c}. \quad (16)$$

In particular this holds for $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$ and $G = \mathbb{Z}, \mathbb{Q}, \mathbb{R}$.

Proof. Here $\text{cf}(G) = \aleph_0$, so the lower bound is $\mathfrak{c}$. Every well-ordered subset $S \subseteq \mathbb{R}$ is countable. Indeed, for each nonmaximal $s \in S$ there is a successor s^+ in S ; choose a rational in (s, s^+) . These intervals are disjoint, producing an injection of S minus a possible maximum into $\mathbb{Q}$. There are at most $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$ countable supports in G , and at most $|K|^{\aleph_0} \leq \mathfrak{c}$ coefficient assignments to any one support. Thus $|K((t^G))| \leq \mathfrak{c}$. $\square$

Remark 5.10 (Why the full strength is not a counting argument). For $G = \mathbb{R}$ and $K = \mathbb{R}$ or $\mathbb{C}$, the base $\mathcal{F}_G(K)$ already has cardinality $\mathfrak{c}$, since it contains K and lies in a field of cardinality $\mathfrak{c}$. Equality of these cardinalities gives no lower bound at all on relative transcendence degree. The separation and coding proof is what supplies it. Nor is an independent family of maximum possible cardinality automatically a transcendence basis: maximal cardinality need not mean maximality under inclusion.

6 Explicit countable witnesses and examples

The coefficient code gives a family of optimal cardinality with integer coefficients, but its indexing uses a set-theoretic bijection. It is useful to have additional closed-form witnesses.

6.1 Factorial gaps with exponential coefficients

Theorem 6.1 (Explicit multiplicative-parameter family). *Let u be an order unit of G and let L contain $\mathbb{R}$. If $q_1, \dots, q_m > 0$ are multiplicatively independent real numbers, then*

$$Z_{q_j} = \sum_{n \geq 2} q_j^n t^{n!u} \quad (1 \leq j \leq m) \quad (17)$$

are algebraically independent over $\mathcal{F}_G(L)$. Consequently the series Z_p , indexed by the ordinary prime numbers p , are jointly algebraically independent, with positive integer coefficients.

Proof. Multiplicative independence means that $\prod_j q_j^{m_j} = 1$ with $m_j \in \mathbb{Z}$ forces every $m_j = 0$. Let $P \in \mathcal{B}_G(L)[X]$ be a nonzero polynomial of positive total degree d . At the vector $c(n) = (q_1^n, \dots, q_m^n)$ its top part is

$$P_d(c(n)) = \sum_{|v|=d} p_v Q_v^n, \quad Q_v = \prod_j q_j^{v_j}.$$

The Q_v are distinct nonzero elements of L . Enumerate the $r \geq 1$ nonzero top coefficients as $p_1, \dots, p_r$ and their associated values as $Q_1, \dots, Q_r$. For any integer $N \geq 2$, the matrix $(Q_i^{N+k})_{0 \leq k < r, 1 \leq i \leq r}$ has determinant

$$\left(\prod_{i=1}^r Q_i^N \right) \prod_{1 \leq i < j \leq r} (Q_j - Q_i) \neq 0.$$

It is therefore invertible over L , and also over $L((t^G))$. Consequently the r values $P_d(c(N+k))$, $0 \leq k < r$, cannot all vanish, since that would force every $p_i = 0$. The factorial gaps satisfy (8) for all sufficiently large n . Choose N beyond that threshold and one nonzero evaluation in the displayed block. Lemma 4.4 proves that P does not vanish on the series. This finite Vandermonde argument treats the Hahn coefficients directly; it requires no limiting argument. Clear denominators for a polynomial over the fraction field. Finally, unique factorization of ordinary integers makes any finite set of distinct primes multiplicatively independent. $\square$

An alternative proof uses Corollary 3.4 to reduce to ordinary Laurent series over $\mathbb{R}(t^u)$. The direct support proof also records the exact obstruction to cancellation over bounded-support Hahn coefficients.

6.2 A continuum family with only integer coefficients

For $\kappa = \aleph_0$ one can choose an effective enumeration of all triples

$$(s, h, \rho), \quad s \subseteq \mathbb{N} \text{ finite}, \quad h : \mathcal{P}(s) \rightarrow \mathbb{N}_{>0}, \quad \rho \in \mathbb{N}.$$

Thus for each binary sequence $A \subseteq \mathbb{N}$, formula (13) gives an integer coefficient stream. Relabeling position $n-2$ by $n \geq 2$ (and writing $c_A(n)$ for the relabeled coefficient),

$$W_A = \sum_{n \geq 2} c_A(n) t^{n!u} \quad (18)$$

is a continuum-sized independent family whenever u is an order unit. Each individual stream is computable relative to the characteristic function of A : evaluating a coefficient asks only finitely many membership questions. The family is not asserted to consist of computable streams; there are only countably many computable descriptions.

More strikingly, for $u = 1$ these witnesses lie in $\mathbb{Q}[[t]]$, yet they remain independent over $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$. All ordinary complex constants and all bounded real-exponent Hahn series have already been admitted in that base. Cofinal descent explains why this does not erase ordinary formal-series transcendence.

6.3 A small catalogue of contrasting behavior

The examples below keep the exponent group explicit.

Series or setting	Consequence
$\sum_{n \geq 0} t^n, G = \mathbb{R}$	Equals $(1 - t)^{-1}$ in $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$, but does not have bounded support.
$\sqrt{1 + t}, G = \mathbb{R}$	Algebraic of degree two over $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$; not a bounded-support fraction.
$\sum_{n \geq 2} t^{n!}, G = \mathbb{R}$	Transcendental over $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$ by Corollary 4.6.
$\sum_{n \geq 2} p^n t^{n!}, p \text{ prime}$	A fully explicit countable independent family over the same base.
$\mathbb{R} \hookrightarrow \mathbb{R}\omega + \mathbb{R} \subset \mathbf{No}$	Every series with real exponents becomes bounded in the larger exponent group.
$\text{cf}(G) > \aleph_0$	Every countable-support Hahn series already belongs to $\mathcal{B}_G(K)$.

The first three rows show why “unbounded support” is not synonymous with “transcendental.” The final two show why a transcendence assertion must name its ambient exponent group and its base field.

7 Rank structure and the unit group

This section concerns the coefficient ring itself. The field criterion is proved from the Hahn field construction. The unit classification additionally uses the known rank-one upper-support theorem, which is isolated explicitly.

7.1 An exact field criterion

For a positive element $a \in G$, define its principal convex subgroup

$$C(a) = \{g \in G : |g| \leq na \text{ for some } n \geq 1\}.$$

It is proper precisely when a is not an order unit.

Theorem 7.1 (Absence of a highest scale). *For any field K and nonzero ordered abelian group G , the following are equivalent:*

1. G has no order unit;
2. $\mathcal{B}_G(K) = \bigcup_{C \subsetneq G \text{ convex}} K((t^C))$;
3. $\mathcal{B}_G(K)$ is a field.

Proof. Assume there is no order unit. If f has support in $[\ell, u]$, choose $a > 0$ with $a \geq |\ell|, |u|$. Then its support lies in the proper convex subgroup $C(a)$, so $f \in K((t^{C(a)}))$. Conversely, a proper convex

subgroup C is bounded above in G : choose a positive $g \notin C$. Every positive $c \in C$ is smaller than g , since otherwise convexity would put g in C . Hence every series in $K((t^C))$ has bounded support in G .

Convex subgroups of an ordered abelian group are linearly ordered by inclusion. To see this, if neither of C, D contained the other, choose positive $c \in C \setminus D$ and $d \in D \setminus C$; whichever is smaller belongs to the other's convex subgroup, a contradiction. Thus the union in (2) is a directed union of fields and is a field.

Finally suppose $u > 0$ is an order unit. The bounded-support element $1 - t^u$ has inverse in the Hahn field

$$(1 - t^u)^{-1} = \sum_{n \geq 0} t^{nu}.$$

This is the usual strongly summable geometric identity. Its support is unbounded because u is an order unit. The inverse is unique in the Hahn field, so no inverse can belong to $\mathcal{B}_G(K)$. Thus (3) excludes an order unit and implies (1). $\square$

Corollary 7.2 (Closed coefficient fields at unbounded rank). *Suppose G is divisible and has no order unit. If K is real closed, then $\mathcal{B}_G(K)$ is real closed. If K is algebraically closed of characteristic zero, then $\mathcal{B}_G(K)$ is algebraically closed.*

Proof. A convex subgroup of a divisible ordered abelian group is divisible: the unique n th part of an element of the subgroup has no larger absolute value, so belongs to the subgroup by convexity. The standard Hahn-field closedness criterion makes each $K((t^C))$ in Theorem 7.1 real closed or algebraically closed, respectively; this is the imported criterion recorded in [1, Fact 2.1.1].

A polynomial has only finitely many coefficients, so all of them lie in one member of the directed union. In the algebraically closed case it has a root in that member. In the real closed case the same argument supplies roots of odd-degree polynomials and square roots of positive elements, which characterize real closedness. The order throughout is the inherited Hahn order. $\square$

These fields can be proper and dense in the full Hahn field. There is no conflict with algebraic closedness: the remaining extension is transcendental. Theorem 5.3 measures how large it is. (Added in batch 32.) For divisible G and $K = \mathbb{R}, \mathbb{C}$ the hypothesis “no order unit” is also necessary: Theorem 14.7 shows that with an order unit u the element $1 - t^u$ has no square root in $\mathcal{F}_G(\mathbb{C})$.

7.2 Reduction to the highest Archimedean quotient

Suppose G has an order unit u . Define

$$C = \{g \in G : n|g| < u \text{ for every positive integer } n\}. \quad (19)$$

This is a convex subgroup. It contains zero and is closed under negation. If $g, h \in C$, applying the defining inequalities at $2n$ gives $2n(|g| + |h|) < 2u$, hence $n|g + h| < u$ for every $n \geq 1$. If $|x| \leq |g|$ with $g \in C$, the same inequalities put x in C . It is proper because $u \notin C$.

For $g \notin C$, some n satisfies $u \leq n|g|$. Since u is an order unit, $|g|$ is one too. A convex subgroup containing such a g contains every element bounded in absolute value by a multiple of $|g|$, so it must be all of G . Thus C is the unique maximal proper convex subgroup. The quotient G/C is nonzero Archimedean: otherwise a positive class dominated by all multiples below another class would generate a nonzero proper convex subgroup of the quotient, whose preimage strictly contains C .

Proposition 7.3 (Top-rank presentation). *If G is divisible and has an order unit, choose an ordered embedding of $H = G/C$ in $\mathbb{R}$ and a $\mathbb{Q}$ -linear section of the quotient map. With $M = K((t^C))$, that choice gives isomorphisms*

$$K((t^G)) \cong M((t^H)), \quad \mathcal{B}_G(K) \cong \mathcal{B}_H(M). \quad (20)$$

The full-field isomorphism is the usual iterated Hahn presentation; the second is its bounded-support restriction.

Proof. Since G and C are $\mathbb{Q}$ -vector spaces, choose a complementary subspace. Order on the quotient dominates order within C , so this identifies G with the lexicographic group $H \oplus C$. Regroup a series according to its H coordinate. Its projected support is well ordered: for any nonempty subset of that projection, take the least original supported exponent whose image lies in the subset; its image is least there. Each fiber, a translated subset of the original support, is well ordered too. Conversely, to find a least element in any nonempty subset of a lexicographic support, first choose its least outer coordinate, then its least inner coordinate in that fiber. Thus well-ordered outer support and well-ordered fibers give exactly a Hahn series on G .

Addition agrees coefficientwise. For multiplication there are finitely many outer pairs at any fixed outer exponent; within each such pair, there are finitely many inner pairs at any fixed inner exponent. These are exactly the original convolution terms. Thus the grouping is a field isomorphism, including inverses by uniqueness. This standard presentation is also recorded in [1, Fact 2.4.2].

If the original support is bounded by g , its H projection is bounded by the image of g . Conversely, if its outer support has upper bound $h \in H$, choose $h' > h$. A lift of h' bounds every fiber, regardless of its internal C support. Thus boundedness is equivalent on the two sides. $\square$

Proposition 7.4 (Imported rank-one support theorem). *If M has characteristic zero and $H \subseteq \mathbb{R}$, then for nonzero $f, g \in \mathcal{B}_H(M)$,*

$$\sup_{\mathbb{R}} \text{supp}(fg) = \sup_{\mathbb{R}} \text{supp}(f) + \sup_{\mathbb{R}} \text{supp}(g).$$

This is the monomial-localized form of [1, Proposition 3.5.1]. Choose $a, b \in H$ bounding the supports of f, g . Then $t^{-a}f, t^{-b}g$ have nonpositive support. Apply the imported one-sided theorem over M to them and translate the real suprema back by a, b ; the translations cancel to give the displayed identity. The one-sided theorem is not reproved or claimed as new here. The supremum is taken in $\mathbb{R}$ and need not be an element of H .

Corollary 7.5 (Units see exactly the highest rank). *Suppose K has characteristic zero and G is divisible with an order unit, and let C be (19). A nonzero $f \in \mathcal{B}_G(K)$ is a unit if and only if its support lies in one coset of C . Equivalently, its image in (20) is a single outer monomial with an arbitrary nonzero coefficient from $K((t^C))$. In particular,*

$$1 + t^g \in \mathcal{B}_G(K)^\times \iff g \in C.$$

Proof. For a nonzero rank-one bounded series define

$$w(f) = \sup_{\mathbb{R}} \text{supp}(f) - \min \text{supp}(f) \geq 0.$$

Proposition 7.4 and the usual valuation identity give $w(fg) = w(f) + w(g)$. If $fg = 1$, the nonnegative widths satisfy $w(f) + w(g) = w(1) = 0$, so $w(f) = 0$. Every exponent of f then lies between its minimum and that same real supremum, forcing singleton support. Conversely, if $f = at^h$ with $a \in M^\times$, its inverse $a^{-1}t^{-h}$ also has singleton support and is bounded. Transfer this statement through (20). For the last claim, if $g \neq 0$ the support is $\{0, g\}$, which lies in one coset precisely when $g \in C$. If $g = 0$, the element is 2, a nonzero scalar in characteristic zero, and the same conclusion holds. $\square$

No arbitrary-rank supremum formula has been assumed. In particular the rank-one width proof should not be repeated in G itself using a guessed Dedekind completion: at higher rank such a formula requires additional structure and need not have the same meaning.

8 Actual surreal and surcomplex realizations

8.1 The normal-form bridge and its scope

Let G be a set-sized ordered additive subgroup of $\mathbf{No}$. Conway normal form supplies the embedding

$$\iota_G : \mathbb{R}((t^G)) \longrightarrow \mathbf{No}, \quad \sum_g r_g t^g \mapsto \sum_g r_g \omega^{-g}. \quad (21)$$

The reversed sign is essential: increasing Hahn exponents correspond to decreasing Conway exponents. We use the standard normal-form isomorphism and its field compatibility, as recorded in [1, Section 2.3] and developed in [2]. This is an imported foundational theorem, not an assertion about convergence of real-indexed sums in the fine surreal topology.

Applying (21) to real and imaginary coefficients gives a field embedding

$$\iota_G^{\mathbb{C}} : \mathbb{C}((t^G)) \longrightarrow \mathbf{No}[i].$$

Indeed, coefficientwise real and imaginary parts have well-ordered supports, their union is well ordered, and complex multiplication agrees with multiplication in the quadratic extension. We henceforth identify these Hahn fields with their concrete images, but continue to name G in every bounded-support base.

8.2 Real-exponent surreal numbers

Define the concrete subfields

$$\mathcal{H}_{\mathbb{R}} = \left\{ \sum_{g \in S} r_g \omega^{-g} : S \subseteq \mathbb{R} \text{ well ordered, } r_g \in \mathbb{R} \right\}, \quad \mathcal{H}_{\mathbb{C}} = \mathcal{H}_{\mathbb{R}}[i].$$

Let $\mathcal{B}_{\mathbb{R}}$ and $\mathcal{B}_{\mathbb{C}}$ require S to be bounded above in $\mathbb{R}$, and put $\mathcal{F}_{\mathbb{R}} = \text{Frac } \mathcal{B}_{\mathbb{R}}$ and $\mathcal{F}_{\mathbb{C}} = \text{Frac } \mathcal{B}_{\mathbb{C}}$.

Theorem 8.1 (A continuum of independent actual infinitesimals). *There is a family of positive infinitesimal surreal numbers*

$$\eta_A = \sum_{n \geq 2} c_A(n) \omega^{-n!} \quad (A \subseteq \mathbb{N}),$$

with positive integer coefficients, algebraically independent even over $\mathcal{F}_{\mathbb{C}}$. Moreover

$$\text{trdeg}_{\mathcal{F}_{\mathbb{R}}} \mathcal{H}_{\mathbb{R}} = \text{trdeg}_{\mathcal{F}_{\mathbb{C}}} \mathcal{H}_{\mathbb{C}} = \mathfrak{c}.$$

The separate explicitly given family

$$\eta_p = \sum_{n \geq 2} p^n \omega^{-n!} \quad (p \text{ an ordinary prime})$$

is also algebraically independent over $\mathcal{F}_{\mathbb{C}}$.

Proof. Use the countable coefficient code and factorial separated system in the complex-coefficient theorem. Its coefficients happen to be positive integers, so the witnesses lie in the real subfield. Their leading term is a positive real multiple of ω^{-2} ; the remaining terms have strictly smaller Conway monomials. Thus they are positive infinitesimals. The exact degrees follow from Corollary 5.9, and the prime family from Theorem 6.1, all transported by (21). $\square$

This is not an assertion of transcendence over all surreals or all surcomplex numbers: the witnesses themselves belong to those proper-class fields. It is a relative algebraic theorem over explicitly specified set-sized subfields.

8.3 Maximal families at every infinite regular cardinal

Let κ be an infinite regular cardinal, regarded as an initial ordinal, and form the ordered additive subgroup of actual surreal numbers

$$G_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q} \omega^\alpha. \quad (22)$$

The direct sum means finite rational linear combinations. Comparison is by the largest exponent with a nonzero coefficient. This group has cardinality κ , is divisible, has cofinality κ , and has no order unit. For the cofinality assertion, any family of fewer than κ finite supports uses a bounded set of indices, by regularity; a larger basis monomial dominates it. Conversely the ω^α are cofinal. The absence of an order unit follows by moving one index past the largest index occurring in a proposed unit.

Let $\mathcal{H}_\kappa^{\mathbb{R}}$ and $\mathcal{H}_\kappa^{\mathbb{C}}$ be the real and complex Hahn fields on this concrete exponent group, interpreted in **No** and **No**[i]. Let $\mathcal{B}_\kappa^{\mathbb{R}}$ and $\mathcal{B}_\kappa^{\mathbb{C}}$ be their bounded-support rings.

Theorem 8.2 (Arbitrarily large, explicit-support surreal families). *For every infinite regular cardinal κ :*

1. $\mathcal{B}_\kappa^{\mathbb{R}}$ is a real closed field and $\mathcal{B}_\kappa^{\mathbb{C}}$ is an algebraically closed field.
2. There are 2^κ positive infinitesimal surreal numbers

$$\eta_A = \sum_{\alpha < \kappa} c_A(\alpha) \omega^{-\omega^{\alpha+1}} \quad (A \subseteq \kappa), \quad (23)$$

all with positive integer coefficients and exactly the displayed common support, algebraically independent over $\mathcal{B}_\kappa^{\mathbb{C}}$.

3. Both exact relative transcendence degrees are 2^κ :

$$\text{trdeg}_{\mathcal{B}_\kappa^{\mathbb{R}}} \mathcal{H}_\kappa^{\mathbb{R}} = \text{trdeg}_{\mathcal{B}_\kappa^{\mathbb{C}}} \mathcal{H}_\kappa^{\mathbb{C}} = 2^\kappa.$$

No element with a support of smaller cardinality is transcendental over its corresponding bounded-support base.

Proof. The first assertion is Corollary 7.2. For the second, the concrete choices

$$a_\alpha = \omega^{\alpha+1}, \quad b_\alpha = \omega^\alpha \quad (\alpha < \kappa)$$

form a separated exponent system in G_κ . Earlier a_β are at most b_α . For every finite n , $a_\alpha > nb_\alpha$, and the b_α are cofinal. Thus the two inequalities in (7) follow exactly as in the uncountable construction, and also hold when $\kappa = \aleph_0$. Apply Theorem 5.3 over $\mathbb{C}$ and then the Conway embedding. Since the

base ring is already a field, its fraction field is itself. The leading term of each number is a positive integer multiple of $\omega^{-\omega}$, proving positivity and infinitesimality.

Finally $|G_\kappa| = \text{cf}(G_\kappa) = \kappa$, while $|\mathbb{R}| = |\mathbb{C}| = \mathfrak{c} \leq 2^\kappa$. Corollary 5.8 supplies the upper bounds as well as the equality. Optimality is Corollary 5.4. $\square$

No continuum hypothesis or generalized continuum hypothesis is used. The regularity assumption on κ in this concrete example is genuine: for a singular initial ordinal the direct sum in (22) has cofinality $\text{cf}(\kappa)$, not κ . The general theorem still applies to that group with its actual cofinality, but it does not give the displayed 2^κ equality by the same argument.

8.4 Enlarging a workspace can erase the base distinction

Consider

$$G = \mathbb{R}\omega + \mathbb{R} \subset \mathbf{No}, \quad H = \mathbb{R} \subset G.$$

Every real number is smaller than ω in this ordered group, so H is not cofinal. All real-exponent Hahn series therefore have support bounded above in G . In particular every witness of Theorem 8.1 belongs to $\mathcal{B}_G(\mathbb{C})$, even though it was transcendental over $\mathcal{F}_{\mathbb{C}}$ in the smaller workspace. This is the concrete collapse in (6), not a failure of field embedding.

At the full class level every set of surreal exponents is bounded above by a surreal number. Consequently the naive class-wide version of “bounded support in $\mathbf{No}$ ” imposes no restriction on a Conway normal form. All the nontrivial statements in this article are deliberately workspace-relative. A set-sized cofinal support in G_κ is not a cofinal subclass of all of $\mathbf{No}$. (Added in batch 32.) Section 16 computes the Galois consequence of such an enlargement: one dominating exponent absorbs the whole old Hahn field, hence an algebraic closure of the old complex fraction field, and Section 17 repeats this along an explicit countable tower in $\mathbf{No}[i]$.

The valuation topology used for density is likewise the topology of the named Hahn workspace. For a constructed family member, the bounded initial Hahn truncations

$$Y_{A, < \beta} = \sum_{\alpha < \beta} c_A(\alpha) t^{a_\alpha} \quad (\beta < \kappa)$$

form a convergent net there: each belongs to $\mathcal{B}_G(K)$, and $v(Y_A - Y_{A, < \beta}) = a_\beta$ is cofinal in G . For $\kappa = \aleph_0$ these are finite partial sums. For uncountable κ they are generally infinite bounded-support series. Indeed the net of all finite subsums does not converge to Y_A then: every finite subset omits some index $n < \omega$, leaving a nonzero coefficient below a_ω , so no such subsum belongs to $\{z : v(z - Y_A) > a_\omega\}$. It cannot converge to any other Hahn series either, since convergence would force every coefficient to equal that of Y_A .

Even the convergent initial-truncation net does not converge in the full surreal fine topology. Choose a surreal exponent θ above the set-sized group G . Every nonzero truncation error has a leading Hahn exponent in G and magnitude greater than $\omega^{-\theta}$, so the fine neighborhood of that radius is never reached. This retains the workspace distinction in the repository’s foundations [4] and its notation guide, under “Topologies and strong summation” [8].

9 Relation to other reports in this collection

This section was added at editorial assembly in commit ed88b8f and its statement comparisons were revisited at b4275b5. It compares statements only; no theorem of another report is used in any proof above, and no theorem here is claimed to follow from, or to imply, one of theirs unless this section says so and gives the reason.

9.1 Tail-spans and differential transcendence

The tail-span report [5] is another source of algebraic independence in Hahn fields in the collection. The two reports answer different base-field problems.

Base fields. Its base is a *full* Hahn field $M((t^\Gamma))$ over a smaller characteristic-zero coefficient field M , and its transcendence comes from the coefficients: by its coefficient-field criterion (`tail:thm:coefficient`) a series with coefficients in an algebraic closure of M is algebraic over $M((t^\Gamma))$ exactly when its coefficients generate a finite extension of M . Here the base $\mathcal{F}_G(K)$ keeps the coefficient field of the witnesses (and in Theorem 3.2 may even enlarge it), and transcendence comes from the support. The actual numbers make the difference concrete. With $t = \omega^{-1}$, the witnesses $\eta_A = \sum_{n \geq 2} c_A(n)t^{n!}$ and η_p of Theorem 8.1 have integer coefficients and integer exponents, so they are *elements* of the tail-span base $B_0 = \mathbb{Q}((t^\mathbb{Q}))$. Conversely $\sqrt{2}$, or the bounded real-coefficient series $\sum_{n \geq 2} \sqrt{p_n} t^{1-1/n}$, lies in $\mathcal{F}_\mathbb{R}(\mathbb{R})$ but not in B_0 . Neither base contains the other, and neither report's independence theorem implies the other's. At uncountable cofinality the two criteria separate further: if $\text{cf}(G) > \aleph_0$ and (γ_n) is a strictly increasing sequence in G , the series $\sum_n \sqrt{p_n} t^{\gamma_n}$ has countable support, so it lies in $\mathcal{B}_G(\overline{\mathbb{Q}} \cap \mathbb{R})$ by Lemma 2.4, while its coefficients generate an infinite extension of $\mathbb{Q}$, so it is transcendental over $\mathbb{Q}((t^G))$ by `tail:thm:coefficient`.

What overlaps. Both reports prove a relative transcendence degree $2^{\aleph_0}$ for set-sized Hahn fields with a countably cofinal exponent group, with explicit positive infinitesimal surreal witnesses, and both obtain the upper bound by the same separate cardinality count rather than from the witnesses: the tail-span report for $\text{trdeg}_{\mathbb{Q}((t^\mathbb{Q}))}(\overline{\mathbb{Q}} \cap \mathbb{R})(t^\mathbb{Q})$ (`tail:thm:surreal`), this report for $\text{trdeg}_{\mathcal{F}_G(K)} K((t^G))$ with $0 \neq G \subseteq \mathbb{R}$ and $|K| \leq \mathfrak{c}$ (Corollary 5.9, Theorem 8.1). Both state that an independent family of maximal cardinality is not thereby a transcendence basis, and both are relative to named set-sized workspaces. Both show that enlarging the exponent group along a noncofinal embedding changes the answer for the same surreal expansion: its `tail:thm:closure` example embeds $\mathbb{Q}$ in lexicographic $\mathbb{Q}^2$ by $q \mapsto (0, q)$, whose image is bounded above by $(1, 0)$, and loses approximability by algebraic elements; the example $\mathbb{R} \subset \mathbb{R}\omega + \mathbb{R}$ of Section 8 makes every old series a bounded-support coefficient and so erases transcendence over the bounded base. The phenomena differ; the common cause is the noncofinal embedding. Finally, both warn that finite or bounded prefix data do not decide algebraicity; the tail-span report proves an undecidability result for computable coefficient streams (`tail:prop:undecidable`), while this report proves no undecidability statement (the remark after Corollary 5.6).

What differs. The tail-span report proves differential independence for the Berarducci–Mantova derivation and for surcomplex analytic functions; this report contains no differential or analytic statement. It classifies all relations among its sums through the cofinite span (`tail:thm:tail`); this report does not classify relations, but its descent theorem transports relation ideals between bases, which has no counterpart there. Its displayed independent family uses almost-disjoint infinite index sets. This is a sufficient construction, not a necessary condition for independence. The unrestricted family $(\xi_A)_{A \subseteq \mathbb{N}_{>0}}$ satisfies the linear relation $\xi_{A \cup B} + \xi_{A \cap B} = \xi_A + \xi_B$ (`tail:rem:indexfamily`). The family (η_A) here is indexed by all subsets and is independent, because c_A is the finite-pattern code of Lemma 5.1, not an indicator function; this is why the witnesses were renamed (Section 1.4). Its displayed actual-surreal differential family uses $\Gamma = \mathbb{Q}$; its coefficient-field and cofinite-span theorems allow arbitrary set-sized ordered value groups. This report proves that $\text{cf}(G)$ is the exact support threshold over the bounded base.

9.2 Entire functions at arbitrary rank

The entire-functions report [6] uses the same invariant, $\text{cf}(G)$, and the same order unit (Section 2.3), for a different object: the ring of strongly entire power series in one variable over $\mathbb{C}((t^\Gamma))$, with Γ divisible in its main theorems. The parallels are these.

- *The same threshold $\aleph_0$.* There, nonpolynomial entire series exist exactly when $\text{cf}(\Gamma) = \aleph_0$ (`ent:cor:cofinality`). Here, a transcendental element with countable support exists exactly when $\text{cf}(G) = \aleph_0$ (Corollary 5.4). Both rest on the fact that a countable subset of a group of uncountable cofinality is bounded above.
- *The order unit.* There, the order unit separates the second and third cases of the cofinality–rank trichotomy (`ent:thm:main`) and decides which restricted series are units (`ent:thm:units`). Here, it decides whether $\mathcal{B}_G(K)$ is a field (Theorem 7.1); the obstruction is the geometric series $(1 - t^u)^{-1} = \sum_n t^{nu}$, whose support is unbounded exactly because u is an order unit.
- *Cofinal and noncofinal enlargement.* There, a nonpolynomial entire series stays entire over a larger group exactly when the old group is cofinal in it (`ent:thm:main-extension(b)`). Here, the descent map (4) is injective exactly when the subgroup is cofinal (Theorem 1.2). These are different statements about different objects, proved independently; this report assumes no divisibility for its descent theorem.
- *GCD rings.* The entire ring there is proved to be a GCD domain. Appendix A concerns the bounded-support Hahn ring, a different ring, and remains conditional on $(\text{GCD}_{\mathbb{R}, \overline{\mathbb{K}}})$; nothing in the entire-functions report supplies that hypothesis.

9.3 Hidden negative directions

The report on hidden negative directions in surcomplex linear algebra [7], placed in the same commit as this one, proves a third kind of independence. Its base $\mathcal{P}_\Gamma$ is the union of the fields $\mathbb{C}((t^H))$ over finitely generated subgroups $H \subseteq \Gamma$ (the Puiseux field when $\Gamma = \mathbb{Q}$). Its binary-prefix almost-disjoint family of infinite prime sets gives tails $h_A = \sum_{p \in A} t^{(1-1/p)\delta}$ independent over that field by fresh prime denominators. This does not assert independence for all infinite prime sets.

For $\Gamma = \mathbb{Q}$, the fields $\mathcal{P}_\mathbb{Q}$ and $\mathcal{F}_\mathbb{Q}(\mathbb{C})$ are incomparable. Each selected h_A has support bounded above by δ , so it belongs to $\mathcal{B}_\mathbb{Q}(\mathbb{C}) \subseteq \mathcal{F}_\mathbb{Q}(\mathbb{C})$, but it lies outside $\mathcal{P}_\mathbb{Q}$. Conversely, the witnesses W_A of (18) with $u = 1$ belong to $\mathbb{C}((t^\mathbb{Z})) \subseteq \mathcal{P}_\mathbb{Q}$ and are transcendental over $\mathcal{F}_\mathbb{Q}(\mathbb{C})$. For groups without an order unit the containment behaves differently: every finitely generated subgroup is bounded above, so $\mathcal{P}_\Gamma \subseteq \mathcal{B}_\Gamma(\mathbb{C})$. Neither the ring nor its fraction field should be confused with a full Hahn coefficient field.

9.4 Foundations and notation

The distinction between the intrinsic valuation topology of a named Hahn workspace and the fine topology of **No**, used in Section 8, is the collection’s standing convention [8]. The Conway normal-form embedding (21) is imported as a standard theorem here; its formal counterpart in the collection is tracked in `docs/NORMAL_FORM_BRIDGE.md` [9].

9.5 Independent surreal copies

This subsection was added in batch 31. The independent-copies report [16] compares its bounded-support section with this report. Its identification of $\mathcal{F}_H(k)$ with the fraction field of the omnific-type ring of series with negative support plus a constant in $\mathbb{Z}$ or $\mathbb{Z}[i]$, for $k = \mathbb{R}, \mathbb{C}$ (`isc:prop:boundedfrac`)

is the localization of Proposition 2.2, with the constant coefficient restricted. Its one-series theorem `isc:thm:cofinalgap` is compared with Corollary 4.6 and Theorem 5.3 after that corollary. Its coset slices are the projections π_r of Lemma 3.1. With them it proves a statement not made here (`isc:thm:sliceddisjoint`): the full Hahn fields on two subgroups are linearly disjoint over the full Hahn field on their intersection, with no cofinality hypothesis. Here the same projections serve descent over a cofinal subgroup (Theorem 3.2). No statement of this report changes.

10 Verification, formalization, and remaining questions

10.1 What was checked computationally

The accompanying code/`verify.py` uses exact rational arithmetic and sparse finite series. It checks the finite convolution identities underlying coset projection, coefficient-functional descent, the auxiliary-polynomial expansion, support-band separation, and the tail bound. It also checks finite-pattern coding, integer-grid detection, factorial-gap inequalities, and explicit multiplicative-parameter examples. Its seed, counts, and outputs are recorded in `data/verification.json`. The delivered run passed 3,163 exact assertions in twelve groups. This is an assertion count, not a count of independently proved theorems. When the report was placed in the collection, the unchanged script was rerun on a copy under Python 3.14.4 and reproduced the delivered record in every field except the recorded Python version (the delivered run used 3.13.5). The countable coefficient code itself is `code/coefficients.py`, which `verify.py` imports.

These checks are regression tests of the finite mechanisms. They are not proofs of the arbitrary-support, transfinite, or cardinal assertions. In particular, a finite enumeration does not verify that a coding fiber has cardinality κ , and a finite truncation does not verify algebraic independence of an infinite family. Those assertions are established by the proofs above. The script uses only the Python standard library and can be run from the report directory with

```
python code/verify.py -output <scratch>/verification-local.json.
```

Its default output is a file `verification.json` in the current directory; an explicit scratch path keeps the delivered record in `data/` unchanged. The delivered verification record is separate from the PDF build record `data/build_audit.json`, which describes the delivered 24-page PDF, not the PDF of this report. (Added in batch 32.) The second source's suite, with its own record, is described in Section 18.7.

10.2 A feasible formalization decomposition

No Lean proof is supplied or claimed. The results nevertheless break into clear interfaces, without a need to formalize all surreal numbers first. The first interface is Hahn subseries extraction and its module identity (2), followed by coefficient-linear maps (3). The second is a bounded-support subring with finite support bounds and an explicit fraction-field embedding. The third is a finite multivariable polynomial expansion lemma giving (10)–(12). These are the main algebraic proof obligations.

Only after those interfaces are complete is it necessary to add the cofinality construction, finite-pattern coding, and cardinal estimates. The actual-surreal consequences are then transported through the Conway-normal-form embedding. Such a transport must be linked to the repository's exact proved normal-form bridge, not merely to an abstract Hahn field with the same informal description. The current state of that bridge in the Lean development is recorded in `docs/NORMAL_FORM_BRIDGE.md` [9]; no statement of this report has an implementation mapping in the collection's formalization

ledger. Its 30 standard result environments are listed in the statement inventory there, which records source coverage rather than Lean proofs. The optional GCD descent in Appendix A belongs to a separate dependency branch.

10.3 Questions left by these theorems

The results do not determine $\text{trdeg}_{\mathcal{F}_G(K)} K((t^G))$ for every large ordered group: the lower bound $2^{\text{cf}(G)}$ can be smaller than the cardinality of the Hahn field, and no general matching upper bound is asserted. For coefficient fields of finite characteristic, the descent and field criterion still hold, and the independent-family argument works with a countably infinite alphabet in an infinite coefficient field. The positive-integer formulation is specific to characteristic zero. Finite coefficient fields require a different coding argument and are not covered by the independence theorem here.

A second question is whether stronger naturality can be imposed on the 2^κ -sized independent family. The existence proof chooses an enumeration of finite tables. It is uniform once that enumeration is fixed, but it is not invariant under every automorphism of the exponent group. The explicit prime family removes that choice at the cost of being only countable.

Finally, the GCD question below concerns divisibility inside the bounded ring, while the main theorem concerns algebraic independence over its fraction field. Neither assertion substitutes for the other. In particular, proving a ring to be GCD would not make the huge transcendental extension disappear.

11 The Galois part: overview, provenance and notation

Sections 11–19 were added in batch 32. They develop the Galois theory of the bounded-support fraction fields $\mathcal{F}_G(\mathbb{C})$ and $\mathcal{F}_G(\mathbb{R})$, using the descent theorem of Section 3 as their engine. Everything in them comes from a second manuscript (source 02) unless it is marked [merge]; a [merge] mark covers the sentence or paragraph it heads and records text written when the manuscript was merged into this report.

An algebraically closed ambient field need not reveal the algebraic complexity of its natural subfields. The surcomplex field $\mathbf{No}[i]$ is an especially striking setting for this distinction. A prescribed exponent group gives a full Hahn field, a canonical omnific subring, and the fraction field of that subring. The first of these is algebraically closed when the coefficients are complex and the exponent group is divisible. The third can have Galois extensions with every finite group. The relevant boundary is not merely the number of available exponents. It is whether a single exponent controls their Archimedean growth, that is, whether G has an order unit (Section 2); equivalently, whether G has a largest Archimedean class. A group can have countable cofinality without having an order unit.

11.1 Source and provenance of this part

Source 02. *A Galois Rank Dichotomy for Omnific Fraction Fields: finite covers, independent radical towers, and collapse at a new surreal scale*, a research manuscript prepared for Vladimir Reshetnikov with ChatGPT, dated 23 September 2026; a 27-page PDF (four front-matter pages and 23 numbered pages). It is manuscript 06 of batch 32 (archive `surreal_galois_rank_article`, main file `galois_rank_dichotomy.tex`), placed as an addition to this report in commit 7d04483 with the local file prefix 02-. Its repository comparison is pinned to 343dc2c471212bb9b53ff4623bace2e1943f255b. At that pin this report consisted of Sections 1–10 and the appendices with the numbering used here, so the source’s citations of Theorem 3.2, Corollaries 3.3–3.4, Theorem 7.1, Corollary 7.2 and Section 8.4

still refer to the same statements. The batch-31 comparisons (the paragraph after Corollary 4.6 and Section 9.5) were added after the pin.

What it contributed. Every result, proof, example, remark, question and non-claim of Sections 11–19: the omnific fraction bridge and complexification; finite coset slices; the transport of whole finite Galois correspondences; realization of every finite group, and of products of at most $\mathfrak{c}$ finite groups, at an order unit; the preserved local Galois envelope; explicit radical towers with exact degrees and group $\prod \widehat{\mathbb{Z}}$; the closedness boundary; an explicit S_m family; real multiquadratic and generalized-dihedral extensions; the cardinality $2^{\mathfrak{c}}$ of the absolute Galois group in real rank; absorption of the old Hahn field by one dominating exponent and the resulting all-or-nothing Galois transition; the explicit omnific quotient $\sqrt{1 - \omega^{-1}} = (\omega^\omega - \dots)/\omega^\omega$; the countable tower in $\mathbf{No}[i]$; twelve research questions.

Where the merge chose. [merge]

1. *Printed once.* Source 02 re-derives, and credits, several results of Sections 1–10. Each is printed once, at its original place, and credited there: its Theorem 3.2 (local-field descent) is Corollary 3.4 with $L = K$; its Theorem 3.5 (cofinal subgroup descent) is Theorem 3.2 with $L = K$; its Proposition 7.1 is Theorem 7.1; the positive half of its Theorem 7.2 is Corollary 7.2; the inclusion of its Theorem 10.1 is (6); its normal-form embedding is (21); and its closing discussion of the full surreal field repeats Section 8.4. Its own proofs of the first two are the coset-projection proof of Theorem 3.2 without the coefficient functional (not needed when $L = K$); they are printed as marked second routes in Remarks 12.5 and 12.8.
2. *Generality.* Source 02 states its Corollary 3.3 over the base $K(s)$, and in its transition theorem applies the same argument over $\mathcal{F}_H(\mathbb{C})$ for cofinal H . Corollary 12.6 is stated at the generality of Theorem 3.2, which contains both uses.
3. *A stale ledger entry.* Source 02’s ledger calls the Galois-group part of its Corollary 3.3 “already implicit in prior degree preservation”. It is explicit: the paragraph after Corollary 3.3 proves preservation of degree and Galois group after compositum. Only the intermediate-field correspondence is added (Section 18.8).
4. *Stale against the tree.* Source 02’s Proposition 2.1 (the fraction bridge, Proposition 12.1) was printed after the pin, for $D = \mathbb{Z}$ or $\mathbb{Z}[i]$, as `isc:prop:boundedfrac` in the independent-copies report [16], which calls it a localization of Proposition 2.2. Source 02 allows any unital subring D and any coefficient field.
5. *Questions.* Source 02’s twelve research directions, printed as subsections there, are Questions 19.1–19.12 here, with their text kept.
6. *Notation.* Source 02’s notation is translated into this report’s (Section 11.2); its two definitions of E_T are one definition here.
7. *Named questions.* Source 02 answers none of the questions of Section 10.3 (transcendence degree for arbitrary G , naturality of the independent family, GCD); none of them is changed.

Numbering. Source 02’s results appear here as follows.

Source 02	This report
Theorems 1.1, 1.2, 1.3	Theorems 11.1, 11.2, 11.3
Propositions 2.1, 2.3; Remark 2.2	Propositions 12.1, 12.3; Remark 12.2
Lemma 3.1; Theorem 3.2; Corollary 3.3	Lemma 12.4; Corollary 3.4 and Remark 12.5; Corollary 12.6
Remark 3.4; Theorem 3.5	Remark 12.7; Theorem 3.2 and Remark 12.8
Lemma 4.1; Theorems 4.2, 4.3	Lemma 13.1; Theorems 13.2, 13.3
Lemma 5.1; Theorem 5.2; Corollary 5.3	Lemma 13.4; Theorem 13.5; Corollary 13.6
Lemma 6.1; Proposition 6.2; Theorem 6.3; Remark 6.4; Theorem 6.5	Lemma 14.1; Proposition 14.2; Theorem 14.3; Remark 14.4; Theorem 14.5
Proposition 7.1; Theorem 7.2; Example 7.3	Theorem 7.1 and Remark 14.6; Theorem 14.7; Example 14.8
Theorem 8.1; Remark 8.2	Theorem 15.1; Remark 15.2
Proposition 9.1; Theorems 9.2, 9.3	Proposition 15.3; Theorems 15.4, 15.5
Theorems 10.1–10.3; Remark 10.4	Theorems 16.1, 16.2, 16.3; Remark 16.4
Proposition 10.5; Remark 10.6	Proposition 16.5; Remark 16.6
Section 11; Theorem 11.1	Section 17; Theorem 17.1
Sections 12, 14; Appendices A, B	Section 18
Section 13 (twelve subsections)	Questions 19.1–19.12

Reading guide (from source 02). The main statements are in Section 11.3. The support proof of Section 12 is the algebraic engine. Readers interested in explicit numbers can proceed from there to Sections 14, 15 and 17. The rank-change theorem of Section 16 is independent of the geometric construction of arbitrary finite groups in Section 13. The questions of Section 19 are proposed directions, not assertions that all have been recognized as open in the published literature.

The delivered audit. Source 02’s audit, shipped as 02-galois-rank-SOURCE_AUDIT.md, records its claim boundary, the repository files it read at the pin, the primary sources it used (L’Innocente–Mantova and Milne’s two sets of course notes; a Wikipedia article was consulted for orientation only, and Conway’s and Gonshor’s books are cited as background without a fresh review), its proof checkpoints, and its computational and build validation. It speaks at the pin and in the delivered file layout.

11.2 Notation of the Galois part

Source 02 writes Γ and k where this report writes G and K , and several of its letters already carry other meanings here. The Galois part uses this report’s notation throughout. [merge] The renamed symbols are these; the tempting false reading is named where there is one.

1. *Groups and fields.* The exponent group Γ is G , and the coefficient field k is K (usually $K = \mathbb{R}$ or $\mathbb{C}$, as stated). Source 02’s calligraphic $\mathcal{K}_\Gamma(k)$ is the *full* Hahn field, written $K((t^G)) = K((t^G))$ here. *False reading:* in this report K is always a coefficient field, never a Hahn field. Source 02’s $\mathcal{B}_\Gamma(k)$ and $\mathcal{F}_\Gamma(k)$ are exactly $\mathcal{B}_G(K)$ and $\mathcal{F}_G(K)$ of (1). The notation guide’s F_Γ is the full field $\mathbb{R}((t^\Gamma))$ (Section 2.3); it is never meant here.
2. *The omnific-type ring.* Source 02’s $\mathcal{A}_\Gamma(D, k) = D + k((t^{\Gamma^{<0}}))$ is $\mathcal{A}_G(D, K) = D + K((t^{G^{<0}}))$, for a unital subring $D \subseteq K$. The independent-copies report writes the case $D = D_K(\mathbb{Z} \text{ or } \mathbb{Z}[i])$ as $\mathcal{A}_K(H)$, with the two arguments in the other order [16]. Source 02’s $P_\Gamma(k) = k((t^{\Gamma^{\leq 0}}))$ and $I_\Gamma(k) = k((t^{\Gamma^{<0}}))$ are written out as $K((t^{G^{\leq 0}}))$ and $K((t^{G^{<0}}))$. *False reading:* source 02’s $I_\Gamma(k)$ is a ring of series, not the relation ideal $I_K(z)$ of Corollary 3.3.

3. *The local coordinate.* Source 02's $z = t^u$, for an order unit u , is $s = t^u$, as in the example after Corollary 3.4; its $\mathbb{C}(z)$, $\mathbb{C}((z))$, $\mathbb{C}[[z]]$ are $\mathbb{C}(s)$, $\mathbb{C}((s)) = \mathbb{C}((t^u))$, $\mathbb{C}[[s]]$, and its curve points $z = a^{-1}$ are $s = a^{-1}$. In Section 3 the letter z denotes a series or tuple in $K((t^H))$.
4. *Extensions and groups.* Source 02's finite extensions L, L_j are Λ, Λ_j : here L is a larger coefficient field (Theorem 3.2). Its finite groups H, H_j are Φ, Φ_j , and its subgroups H of a Galois group are Ψ : here H is an exponent subgroup. Its absolute Galois group G_E is $\mathfrak{G}_E$: here G is the exponent group.
5. *Inclusions of exponent groups.* Source 02's inclusion $\Gamma \subseteq \Delta$ (its Section 10) is $H \subseteq G$, the small group first, as in Theorem 3.2 and (6); its cofinal subgroup $H \subseteq \Gamma$ (its Theorem 3.5) is $H \subseteq G$. Its dominating exponent h is θ , as in Section 8.4, where θ is already a surreal exponent above G : here h is a series in $f = at^g(1 + h)$ and an element of H . Its lexicographic $\mathbb{Q}h \oplus \Gamma$ is $\mathbb{Q}\theta \oplus H$.
6. *Convex subgroups and the tower.* Source 02's principal convex subgroup Γ_g is $C(g)$ of Section 7. Its tower Γ_n, Γ_∞ (which clashes with Γ_g in the source) is Ω_n, Ω_∞ ; the group G_κ of (22) is a different group. Its F_n^k, K_n^k are $\mathcal{F}_{\Omega_n}(K)$ and $K((t^{\Omega_n}))$.
7. *Actual fields.* Source 02's $F_0^{\mathbb{R}} = \mathcal{F}_{\mathbb{R}}(\mathbb{R})$ and $F_0^{\mathbb{C}} = \mathcal{F}_{\mathbb{R}}(\mathbb{C})$ are $\mathcal{F}_{\mathbb{R}}(\mathbb{R})$ and $\mathcal{F}_{\mathbb{R}}(\mathbb{C})$, identified by (21) with $\mathcal{F}_{\mathbb{R}}$ and $\mathcal{F}_{\mathbb{C}}$ of Section 8.2.
8. *The local envelope.* Source 02's $\mathcal{U}_0$ and $\mathcal{U}_\Gamma$ are $\mathcal{E}_0$ and $\mathcal{E}_G$: here $U(f, \delta)$ is a valuation ball (Corollary 5.6).
9. *One field E_T .* Source 02 defines $E_T = \mathcal{F}_\Gamma(\mathbb{C})(r_{a,n} : a \in T, n \geq 1)$ twice, for $T \subseteq \mathbb{C}^\times$ and again for $T \subseteq \mathbb{R}_{>0}$; the second is the first restricted to real positive T . Here it is defined once, in (14.3).
10. *Local letters.* In proofs, F, F' and F_0 abbreviate the fraction fields named at each use; source 02's compositum M is $\Lambda_{<s}$ (here M is a coefficient field in Proposition 7.3), its field $K = \mathcal{K}_\Gamma(\mathbb{C})$ in the splitting proof is $\mathbb{C}((t^H))$, and its map $f(x) = x^m - x$ is φ . Where source 02 counts radicals or extensions by s (its Theorems 1.2 and 6.3, Lemma 5.1), the count is q here, since s is the local coordinate.

Unchanged: the order unit u , the sign convention $t^\gamma \leftrightarrow \omega^{-\gamma}$ of (21), the valuation $v, \mathfrak{c} = 2^{\aleph_0} = |\mathbb{C}|$ (no continuum hypothesis), the radicals $r_{a,n}$, the roots of unity $\zeta_n, \widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z}$ with its profinite topology, and $\mathbf{No}[i]$ identified with complex-coefficient Conway normal forms.

Upper boundedness of a Hahn support must not be confused with bounded cardinality, bounded birthday, or boundedness of the value of the series in the ordered field. Source 02's arguments use ordinary ZFC for set-sized fields. Only the final interpretation of all surreals uses the customary class notation, always with set-sized normal-form supports. The strictly negative t -exponents form the purely infinite part of an omnific integer.

11.3 The principal theorems of the Galois part

Theorem 11.1 (Galois rank dichotomy). *Let G be a nonzero divisible ordered abelian group.*

1. *If G has no order unit, $\mathcal{F}_G(\mathbb{C})$ is algebraically closed and $\mathcal{F}_G(\mathbb{R})$ is real closed.*
2. *If G has an order unit, then for every family $(\Phi_j)_{j \in J}$ of finite groups with $|J| \leq \mathfrak{c}$ there is a Galois extension $E/\mathcal{F}_G(\mathbb{C})$ inside $\mathbb{C}((t^G))$ such that*

$$\mathrm{Gal}(E/\mathcal{F}_G(\mathbb{C})) \cong \prod_{j \in J} \Phi_j$$

as topological groups. In particular, every finite group is realized.

3. In the second case neither $\mathcal{F}_G(\mathbb{C})$ nor $\mathcal{F}_G(\mathbb{R})$ has the corresponding closedness property in (1). Indeed, for an order unit u , $1 - t^u$ has no square root in either fraction field, although its root with constant term 1 belongs to the ambient real Hahn field.

Part (1) is prior material of this report (Corollary 7.2), and the obstruction in (3) follows quickly from its descent theorem. [merge] The example after Corollary 3.4 proves the same obstruction for $1 + t^u$ by the same argument. The additional content developed here is the Galois realization in (2), its simultaneous and explicit variants, and its behavior under rank enlargement. The proofs of all three parts are supplied, with the old ingredients explicitly identified: (1) and (3) in Theorem 14.7, (2) in Theorem 13.5.

Theorem 11.2 (Explicit independent radical towers). *Suppose u is an order unit of a nonzero ordered abelian group G , and put $s = t^u$. For $a \in \mathbb{C}^\times$ and $n \geq 1$, let*

$$r_{a,n} = \sum_{m \geq 0} \binom{1/n}{m} (-a)^m s^m.$$

For distinct $a_1, \dots, a_q$ and $n \geq 2$,

$$[\mathcal{F}_G(\mathbb{C})(r_{a_1,n}, \dots, r_{a_q,n}) : \mathcal{F}_G(\mathbb{C})] = n^q,$$

and the Galois group is $(\mathbb{Z}/n\mathbb{Z})^q$. The field generated by all $r_{a,n}$ has Galois group $\prod_{a \in \mathbb{C}^\times} \widehat{\mathbb{Z}}$. Divisibility of G is not required for these assertions.

[merge] The number of radicals is written q here, and in Theorem 14.3, because s is the local coordinate; source 02 writes s . The theorem is proved in Theorems 14.3 and 14.5.

The second principal phenomenon goes in the opposite direction. One new scale absorbs not just a selected root, but an algebraic closure of the entire old complex fraction field.

Theorem 11.3 (One-rank absorption and renewed obstruction). *Let H be nonzero and divisible, and form $G = \mathbb{Q}\theta \oplus H$ with the lexicographic order in which θ dominates H . Then*

$$K((t^H)) \subseteq \mathcal{B}_G(K) \subseteq \mathcal{F}_G(K) \quad (K = \mathbb{R}, \mathbb{C}).$$

The field $\mathcal{F}_G(\mathbb{C})$ contains an algebraic closure of $\mathcal{F}_H(\mathbb{C})$. For every finite extension $\Lambda/\mathcal{F}_H(\mathbb{C})$,

$$\Lambda \otimes_{\mathcal{F}_H(\mathbb{C})} \mathcal{F}_G(\mathbb{C}) \cong \mathcal{F}_G(\mathbb{C})^{[\Lambda:\mathcal{F}_H(\mathbb{C})]}.$$

Nevertheless, θ is an order unit of G , so $\mathcal{F}_G(\mathbb{C})$ again has the realization property of Theorem 11.1(2). Its own element $1 - t^\theta$ has no square root in $\mathcal{F}_G(\mathbb{C})$.

It is proved at the end of Section 16. More generally, Theorem 16.3 proves an exhaustive alternative: cofinal exponent-group inclusions preserve every finite extension and its degree, while noncofinal inclusions split every finite extension of the old complex fraction field. On absolute Galois groups, the induced restriction is respectively surjective or trivial.

This theorem is about *extension of the scalar field*. It does not say that a previously nontrivial field extension becomes equal to its base, or that a Galois automorphism ceases to exist as an abstract map. Rather, its finite separable algebra becomes a product of copies of the new base. This distinction will be important in Section 16.

11.4 Relationship with Sections 1–10 and with published work

Section 3 proves cofinal linear disjointness (Theorem 3.2), including the special case

$$K((s)) \text{ and } \mathcal{F}_G(K) \text{ are linearly disjoint over } K(s) \quad \text{when } s = t^u \text{ and } u \text{ is an order unit} \quad (11.1)$$

(Corollary 3.4). Section 7 proves that the bounded-support ring is a field exactly when there is no order unit, and its real or algebraic closedness under the corresponding hypotheses. Those are not new claims of the Galois part. Source 02 gives a short direct proof of (11.1) that is sufficient for all the Galois applications (Remark 12.5).

The normal-form and Hahn-field background is classical; the formulation used is that of L’Innocente and Mantova [1, Sections 2.1, 2.3–2.4]. Finite covers of the punctured complex sphere are supplied by the Riemann existence theorem, in the form of Milne [14, Theorem 21.3]. Elementary finite and infinite Galois theory can be found in [13]. None of these ingredients is claimed as an invention.

The proposed contribution of source 02 is a specific synthesis: local algebraic branches survive unchanged over omnific fraction fields at an order-unit scale; independent finite covers give arbitrarily prescribed finite-group products; explicit radical towers record a large profinite quotient; and a single dominating scale simultaneously splits every old finite extension. Its targeted source inspection did not establish historical priority for this package. In particular, no implication is made that a standard consequence has never been noticed in another language.

12 Omnific fractions and cyclic-coset descent

12.1 Supports, summation, and the ambient field

Source 02’s Hahn-series conventions are those of Section 2: well-ordered supports, coefficientwise addition, convolution, the valuation $v(f) = \min \text{supp}(f)$ with $v(0) = +\infty$, and the field $K((t^G))$ [1, Fact 2.1.1]. When K is ordered, the sign of $f \neq 0$ is the sign of its leading coefficient. Only the usual Hahn valuation is used, not an ordinal-valued support degree.

No analytic convergence is being used. For example, if $u > 0$, the family $(c_m t^{mu})_{m \geq 0}$ is a legitimate Hahn family because $\{mu : m \geq 0\}$ is well ordered and its terms have distinct exponents. Its sum exists even when the family is bounded above in a larger ordered group. Neither ordinary complex convergence nor convergence in a fine topology on all of **No** is needed.

12.2 From an omnific subring to bounded fractions

The set $\mathcal{B}_G(K)$ is a ring (Lemma 2.1). For a unital subring $D \subseteq K$ put

$$\mathcal{A}_G(D, K) = D + K((t^{G^{<0}})) \subseteq K((t^{G^{\leq 0}})) = K + K((t^{G^{<0}})).$$

It is a subring. For $(D, K) = (\mathbb{Z}, \mathbb{R})$ it is the canonical omnific subring at scale G , and for $(D, K) = (\mathbb{Z}[i], \mathbb{C})$ its Gaussian counterpart.

Proposition 12.1 (The fraction-field bridge). *For every nonzero ordered abelian group G , every field K and every unital subring $D \subseteq K$,*

$$\text{Frac } \mathcal{A}_G(D, K) = \text{Frac } K((t^{G^{\leq 0}})) = \text{Frac } \mathcal{B}_G(K) = \mathcal{F}_G(K) \quad (12.1)$$

inside $K((t^G))$. In particular the field is independent of D .

Proof. The inclusions of rings from left to right are immediate. Conversely, let $f \in \mathcal{B}_G(K)$, with support bounded above by b . Choose $h > 0$ with $h > b$. Such an h exists in every nonzero ordered abelian group: add a positive element to $\max(b, 0)$. Then

$$t^{-h}f \in K((t^{G^{<0}})), \quad t^{-h} \in K((t^{G^{<0}})) \setminus \{0\}, \quad f = \frac{t^{-h}f}{t^{-h}}.$$

Thus $\mathcal{B}_G(K) \subseteq \text{Frac } \mathcal{A}_G(D, K)$, proving the reverse inclusion after taking fractions. The argument also covers constant coefficients in K that do not belong to D . $\square$

[merge] This is the localization of Proposition 2.2 (which writes $\mathcal{B}_G(K) = t^G K((t^{G^{\leq 0}}))$), with the constant coefficient restricted to D ; the independent-copies report prints the case $D = \mathbb{Z}, \mathbb{Z}[i]$ as `isc:prop:boundedfrac` [16]. The bridge is why the Galois part concerns omnific arithmetic rather than an unrelated subfield construction.

Remark 12.2. The equality (12.1) is not the assertion $\text{Frac } \mathcal{A}_G(D, K) = K((t^G))$. That stronger equality frequently fails. The distinction is central to the Galois part and to the cofinality discussion in [1, Proposition 2.4.5]; Theorem 5.3 measures the gap.

Proposition 12.3 (Complexification). *For every nonzero ordered abelian group G ,*

$$\mathcal{B}_G(\mathbb{C}) = \mathcal{B}_G(\mathbb{R}) + i \mathcal{B}_G(\mathbb{R}), \quad \mathcal{F}_G(\mathbb{C}) = \mathcal{F}_G(\mathbb{R})(i).$$

The fixed field of coefficientwise conjugation on $\mathcal{F}_G(\mathbb{C})$ is $\mathcal{F}_G(\mathbb{R})$.

Proof. Real and imaginary coefficient parts have supports contained in the original support. This proves the first identity. If a fraction has numerator $a + ib$ and denominator $c + id$, with all four terms real and bounded supported, multiplication by $c - id$ expresses it in $\mathcal{F}_G(\mathbb{R})(i)$. The denominator $c^2 + d^2$ is nonzero because $\mathbb{R}((t^G))$ is ordered. The reverse inclusion and the fixed-field assertion follow at once. $\square$

The basic closedness input. Source 02 uses the classical result that $K((t^G))$ is algebraically closed when K is algebraically closed and G is divisible, and is real closed when K is real closed and G is divisible [1, Fact 2.1.1], as Corollary 7.2 does. Its use is restricted to those places where a full Hahn field, not a bounded-support fraction field, appears. The finite Galois constructions in an order-unit group do not themselves require that the ambient Hahn field be algebraically closed.

12.3 Cyclic-coset descent at an order unit

This subsection reproves the cyclic instance of the cofinal linear-disjointness theorem (Theorem 3.2 and Corollary 3.4). The proof explains exactly why an order unit preserves local algebraic obstructions in the larger fraction field.

Lemma 12.4 (Finite slices). *Suppose u is an order unit of G , and $S \subseteq G$ is well ordered and bounded above. For each coset $c + \mathbb{Z}u$, the set $S \cap (c + \mathbb{Z}u)$ is finite.*

Proof. If S is nonempty, write $a = \min S$ and let b be an upper bound. Choose an integer $N > 0$ such that both $|a - c|$ and $|b - c|$ are at most Nu . Any integer m for which $c + mu \in S$ then satisfies $-N \leq m \leq N$. There are only finitely many such integers. $\square$

In particular, the projection of $f \in \mathcal{B}_G(K)$ onto a single coset is $t^c p_c(s)$ for a Laurent polynomial $p_c \in K[s, s^{-1}]$, where $s = t^u$. In the notation of (2) with $H = \mathbb{Z}u$, $t^c p_c(s)$ is $t^c \pi_c(f)$. There can be infinitely many nonzero cosets. Their sum is not replaced by a finite decomposition; only one coset at a time is tested.

Source 02's Theorem 3.2 (local-field descent) states: *if u is an order unit of G and $s = t^u$, then $K((s))$ and $\mathcal{F}_G(K)$ are linearly disjoint over $K(s)$ inside $K((t^G))$, and consequently $K((s)) \cap \mathcal{F}_G(K) = K(s)$.* This is Corollary 3.4 with $L = K$; it is printed once, there.

Remark 12.5 (Second route to Corollary 3.4 for $L = K$, from source 02). Let $x_1, \dots, x_r \in K((s))$ be linearly independent over $K(s)$, and suppose $\sum_{j=1}^r a_j x_j = 0$ with $a_j \in \mathcal{F}_G(K)$. Choose a common nonzero denominator in $\mathcal{B}_G(K)$. After multiplying through, we may assume $a_j \in \mathcal{B}_G(K)$. Fix a representative c of a coset modulo $\mathbb{Z}u$. By Lemma 12.4, the $c + \mathbb{Z}u$ component of a_j is $t^c p_{j,c}(s)$ with $p_{j,c} \in K[s, s^{-1}]$. Every exponent of x_j belongs to $\mathbb{Z}u$, so multiplication by x_j preserves the coset of an exponent of a_j . Projecting the relation onto $c + \mathbb{Z}u$ therefore gives $t^c \sum_{j=1}^r p_{j,c}(s) x_j = 0$. The convolution coefficients are finite Hahn sums, so this projection commutes with the finite sum and with these products. Linear independence over $K(s)$ yields $p_{j,c} = 0$ for every j . This is true for every coset, hence every a_j vanishes. This proves linear disjointness. For the intersection, an element $x \in K((s)) \setminus K(s)$ makes $1, x$ linearly independent over $K(s)$. It therefore cannot belong to $\mathcal{F}_G(K)$.

[merge] This is the mechanism of the proof of Theorem 3.2 (Lemma 3.1), specialized to $H = \mathbb{Z}u$ and $L = K$: no coefficient functional is needed, every coset is projected instead of one chosen exponent, and the projected coefficients are Laurent polynomials by Lemma 12.4.

Corollary 12.6 (Exact preservation of algebraic data). *Let $K \subseteq L$ be fields, let $0 \neq H \subseteq G$ be cofinal, and put $F_0 = \mathcal{F}_H(K)$ and $F = \mathcal{F}_G(L)$. Let Λ/F_0 be a finite extension contained in $K((t^H))$. Then*

$$F \otimes_{F_0} \Lambda \cong F\Lambda, \quad [F\Lambda : F] = [\Lambda : F_0].$$

Minimal polynomials of elements of Λ over F_0 remain minimal after passing to F . If Λ/F_0 is Galois, restriction induces an isomorphism

$$\text{Gal}(F\Lambda/F) \cong \text{Gal}(\Lambda/F_0).$$

The isomorphism preserves the entire subgroup/intermediate-field correspondence: a subgroup Ψ corresponds to $F\Lambda^\Psi$. In particular, for an order unit u of G , $s = t^u$, $H = \mathbb{Z}u$ and $L = K$, this applies to every finite extension $\Lambda/K(s)$ embedded in $K((s))$.

Proof. Linear disjointness (Theorem 3.2) makes the multiplication map on the tensor product injective. Its image is a finite-dimensional domain over F , hence a field and therefore $F\Lambda$. A basis of Λ remains independent, proving the degree assertion. Apply the same argument to $F_0(\alpha)$ to preserve the minimal polynomial of any $\alpha \in \Lambda$.

In the Galois case each automorphism of Λ/F_0 extends by the identity on F to the tensor product, and hence to $F\Lambda$. The resulting $[\Lambda : F_0]$ automorphisms exhaust the automorphisms of the extension. For a subgroup Ψ , $F\Lambda^\Psi$ is fixed by Ψ , and its degree is the correct index by the first part. Galois correspondence gives equality with the fixed field. $\square$

[merge] Source 02 states this corollary for $H = \mathbb{Z}u$ and $L = K$ (base $K(s)$), and applies the same argument with base $\mathcal{F}_H(\mathbb{C})$ in Theorem 16.3; the proof uses only linear disjointness, which Theorem 3.2 supplies in the stated generality. The degree, minimal-polynomial and Galois-group assertions are Corollary 3.3(4) and the paragraph after it; the intermediate-field correspondence is the addition.

Remark 12.7 (Why intersections alone are not enough). The elementary equality $K((s)) \cap \mathcal{F}_G(K) = K(s)$ rules out individual rational representations of a local branch. For simultaneous extensions, one needs the stronger linear-disjointness statement. A trivial field intersection, without appropriate hypotheses, is not a general substitute for linear disjointness. All subsequent degree and Galois assertions use linear disjointness, not merely the intersection.

12.4 The general cofinal subgroup form

An ordered subgroup $H \subseteq G$ is cofinal if every $g \in G$ is at most some $h \in H$ (Section 2). The cyclic subgroup $\mathbb{Z}u$ is cofinal exactly when u is an order unit. For rank transitions source 02 needs the general theorem, which is Theorem 3.2; it reproves its same-coefficient-field case ($L = K$) rather than relying on an unaudited manuscript proof. Its statement (source 02, Theorem 3.5): *for $0 \neq H \subseteq G$ cofinal and any field K , $K((t^H))$ and $\mathcal{F}_G(K)$ are linearly disjoint over $\mathcal{F}_H(K)$ inside $K((t^G))$* . It is printed once, as Theorem 3.2.

Remark 12.8 (Second route to Theorem 3.2 for $L = K$, from source 02). Let $x_1, \dots, x_r \in K((t^H))$ be linearly independent over $\mathcal{F}_H(K)$. In a relation $\sum_j a_j x_j = 0$ with $a_j \in \mathcal{F}_G(K)$, clear a common bounded-support denominator, so that $a_j \in \mathcal{B}_G(K)$. For a coset $c + H$, let $b_{j,c}$ be t^{-c} times the projection of a_j to that coset. It is a Hahn series supported in H . If $\text{supp}(a_j) \leq d_j$ in G , then $\text{supp}(b_{j,c}) \leq d_j - c$. Cofinality supplies an element of H above $d_j - c$, so $b_{j,c} \in \mathcal{B}_H(K) \subseteq \mathcal{F}_H(K)$. Projecting the relation gives $\sum_j b_{j,c} x_j = 0$ in $K((t^H))$. Independence over $\mathcal{F}_H(K)$ implies that every $b_{j,c}$ is zero. Every coefficient of every a_j lies in one of these cosets, so all a_j are zero.

For $H = \mathbb{Z}u$, every upper bounded well-ordered support in H is finite, so $\mathcal{B}_H(K) = K[s, s^{-1}]$ and $\mathcal{F}_H(K) = K(s)$. Thus this proof specializes exactly to the cyclic-coset argument of Remark 12.5. The general theorem adds no novelty claim, but makes the all-extension transition theorem (Theorem 16.3) exhaustive. [merge] The proof is that of Theorem 3.2 with the coefficient functional omitted.

13 Finite covers and prescribed products

13.1 A classical geometric input with a split local place

Here and below a branch point refers to the usual ramification of a finite cover of the complex projective line. It does not refer to the Hahn valuation on $\mathbb{C}((t^G))$.

Lemma 13.1 (Finite groups in a fixed local coordinate). *For every finite group Φ , there is a finite Galois extension $\Lambda/\mathbb{C}(s)$ with group Φ , unramified at $s = 0$ and at infinity, whose entire field embeds over $\mathbb{C}(s)$ into $\mathbb{C}((s))$. Its possible branch points can be confined to any sufficiently large prescribed finite subset of $\mathbb{C}^\times$.*

Proof. The trivial group is immediate. For nontrivial Φ , choose generators $g_1, \dots, g_r$ and append $(g_1 \cdots g_r)^{-1}$. The resulting tuple generates Φ and has product 1. Add identity entries if more marked points have been prescribed.

For a finite set $S \subset \mathbb{C}^\times$ of the required size, $\pi_1(\mathbb{P}^1(\mathbb{C}) \setminus S, 0)$ has loops around the points of S as generators with their product equal to 1. Map these loops to the tuple above. The kernel defines a connected regular topological cover with deck group Φ . The Riemann existence theorem [14, Theorem 21.3] algebraizes this finite cover. Its smooth projective completion is branched only in S . In particular neither 0 nor infinity is a branch point.

Choose a point above 0. Its residue field is $\mathbb{C}$, and its ramification index is 1. The completed local ring is thus $\mathbb{C}[[s]]$, with the base coordinate s itself a uniformizer. The injection of the function field into the fraction field of that completion gives the required embedding $\Lambda \hookrightarrow \mathbb{C}((s))$. Since the cover

is regular, $\Lambda/\mathbb{C}(s)$ is Galois with group Φ . This embeds the whole splitting field, not merely one root of a polynomial. $\square$

Theorem 13.2 (Finite-group realization). *If G has an order unit, every finite group is the Galois group of a finite extension of $\mathcal{F}_G(\mathbb{C})$ inside $\mathbb{C}((t^G))$. The extension has a basis consisting of local Laurent series in a single monomial $s = t^u$.*

Proof. Choose $\Lambda/\mathbb{C}(s)$ from Lemma 13.1, view it inside $\mathbb{C}((s)) \subseteq \mathbb{C}((t^G))$, and apply Corollary 12.6. A $\mathbb{C}(s)$ -basis of Λ remains a basis after scalar extension. $\square$

This is an inverse-Galois statement over the specifically defined field $\mathcal{F}_G(\mathbb{C})$. It says nothing about the inverse Galois problem over $\mathbb{Q}$. The source of arbitrary finite groups is the classical complex covering theorem; the new arithmetic issue is whether their function-field degrees and symmetries survive the much larger omnific fraction field.

13.2 All locally split finite Galois covers at once

Inside the fixed field $\mathbb{C}((s))$, let $\mathcal{E}_0$ be the compositum of all finite Galois extensions of $\mathbb{C}(s)$ that it contains. This is an algebraic Galois extension: every finite collection is contained in a finite Galois compositum. For an order unit u , set

$$\mathcal{E}_G = \mathcal{F}_G(\mathbb{C}) \mathcal{E}_0 \subseteq \mathbb{C}((t^G)).$$

Theorem 13.3 (Galois-envelope preservation). *The multiplication map identifies $\mathcal{F}_G(\mathbb{C}) \otimes_{\mathbb{C}(s)} \mathcal{E}_0$ with $\mathcal{E}_G$, and restriction induces a topological isomorphism*

$$\mathrm{Gal}(\mathcal{E}_G/\mathcal{F}_G(\mathbb{C})) \cong \mathrm{Gal}(\mathcal{E}_0/\mathbb{C}(s)).$$

Every finite Galois diagram in this local envelope, including its restriction maps and its intermediate fields, is preserved.

Proof. Apply Corollary 12.6 to every finite Galois subextension of $\mathcal{E}_0$. The extensions of automorphisms are uniquely determined by acting trivially on $\mathcal{F}_G(\mathbb{C})$, so they commute with restriction between finite subextensions. Passing to the directed union of the fields and the inverse limit of their finite Galois groups proves the assertion. This also proves the topological claim, because both Krull topologies are the inverse-limit topologies of those finite quotients [13, Chapter 7]. $\square$

The theorem is stronger than a disconnected list of finite-group realizations. It preserves the relations among all the covers in the chosen envelope. It does not identify this envelope with an algebraic closure of $\mathcal{F}_G(\mathbb{C})$, or identify its Galois group with the full absolute Galois group. Those additional assertions would need new arguments.

13.3 Independent products of arbitrary finite groups

Lemma 13.4 (Separation by disjoint branch loci). *Let $\Lambda_1, \dots, \Lambda_q$ be finite Galois extensions of $\mathbb{C}(s)$. Suppose their branch loci on $\mathbb{P}^1(\mathbb{C})$ are pairwise disjoint. Then they are jointly linearly disjoint, and*

$$\mathrm{Gal}(\Lambda_1 \cdots \Lambda_q/\mathbb{C}(s)) \cong \prod_{j=1}^q \mathrm{Gal}(\Lambda_j/\mathbb{C}(s)).$$

Proof. We argue by induction on q . The compositum $\Lambda_{<q} = \Lambda_1 \cdots \Lambda_{q-1}$ is Galois and is unramified outside the union of the earlier branch loci. This last assertion can be checked locally: a compositum of finite unramified covers is unramified, and over a complex point an unramified completion is simply the base Laurent series field.

The intersection $\Lambda_q \cap \Lambda_{<q}$ is a finite Galois extension of $\mathbb{C}(s)$. Its branch locus is contained in that of Λ_q and also in that of $\Lambda_{<q}$, hence is empty. A connected finite unramified cover $Y \rightarrow \mathbb{P}_{\mathbb{C}}^1$ of degree d must have $d = 1$: the Riemann–Hurwitz formula gives $2g(Y) - 2 = -2d$, which is impossible for $d > 1$. Therefore $\Lambda_q \cap \Lambda_{<q} = \mathbb{C}(s)$. For these finite Galois extensions, the restriction formula for a compositum gives $[\Lambda_q \Lambda_{<q} : \Lambda_{<q}] = [\Lambda_q : \mathbb{C}(s)]$. Induction proves the product assertion and joint linear disjointness. $\square$

[merge] The number of extensions is written q here and $\Lambda_{<q}$ is source 02's M (Section 11.2).

Theorem 13.5 (Continuum-sized prescribed products). *Suppose u is an order unit of G . For every family $(\Phi_j)_{j \in J}$ of finite groups with $|J| \leq \mathfrak{c}$, there is a Galois extension $E/\mathcal{F}_G(\mathbb{C})$ in $\mathbb{C}((t^G))$ such that*

$$\mathrm{Gal}(E/\mathcal{F}_G(\mathbb{C})) \cong \prod_{j \in J} \Phi_j.$$

One can choose finite Galois subextensions E_j with the specified groups that are jointly linearly disjoint over $\mathcal{F}_G(\mathbb{C})$, and take E to be their compositum.

Proof. Choose pairwise disjoint finite subsets $S_j \subset \mathbb{C}^\times$, each large enough for the construction in Lemma 13.1. This is possible since at most $\mathfrak{c}$ finite sets are required. For example, partition a set of cardinality $\mathfrak{c}$ into $\mathfrak{c}$ countable sets and select the required finite numbers of points.

Construct $\Lambda_j/\mathbb{C}(s)$ with group Φ_j , branched only in S_j , and embed each Λ_j into the same $\mathbb{C}((s))$ by choosing a point above 0. Every finite collection is jointly linearly disjoint by Lemma 13.4. No common branch point at infinity has been introduced: infinity was excluded from all branch loci in Lemma 13.1.

Put $F = \mathcal{F}_G(\mathbb{C})$ and $E_j = F\Lambda_j$. The full finite compositum $\prod_{j \in J_0} \Lambda_j$ lies in $\mathbb{C}((s))$ for every finite $J_0 \subseteq J$. Corollary 12.6 preserves its degree and Galois group, so the E_j are also jointly linearly disjoint. The compositum E is a Galois algebraic extension of F , and

$$\mathrm{Gal}(E/F) = \varprojlim_{J_0 \subseteq J \text{ finite}} \prod_{j \in J_0} \Phi_j = \prod_{j \in J} \Phi_j$$

with the product topology. $\square$

Corollary 13.6. *For an order-unit group G , every product in Theorem 13.5 is a continuous quotient of $\mathfrak{G}_{\mathcal{F}_G(\mathbb{C})}$.*

Proof. Embed the algebraic extension E/F in an algebraic closure of F . Restriction from the absolute Galois group onto the Galois group of a Galois subextension is surjective. $\square$

The conclusion is a quotient assertion. A profinite group can have all finite groups as quotients and still have substantial relations or cohomological restrictions. In particular, the theorem is not a proof that $\mathfrak{G}_{\mathcal{F}_G(\mathbb{C})}$ is a free profinite group.

14 Radical towers and the closedness boundary

The construction of Section 13 is geometric. This section supplies concrete normal forms and an elementary independence argument. It also avoids the need to infer multiradical independence from single-root examples.

14.1 Normalized binomial branches

For $a \in \mathbb{C}^\times$ and $n \geq 1$, define

$$r_{a,n}(s) = \sum_{m \geq 0} \binom{1/n}{m} (-a)^m s^m, \quad \binom{x}{m} = \frac{x(x-1) \cdots (x-m+1)}{m!}. \quad (14.1)$$

These are formal power series, hence Hahn series at $s = t^u$ for any $u > 0$.

Lemma 14.1. *The series $r_{a,n}$ is the unique element of $1 + s\mathbb{C}[[s]]$ satisfying $r_{a,n}^n = 1 - as$. Moreover*

$$r_{a,mn}^m = r_{a,n}. \quad (14.2)$$

Proof. In a formal series $r = 1 + \sum_{j \geq 1} b_j s^j$, the coefficient of s^j in r^n is nb_j plus an expression in the preceding coefficients. Since the characteristic is zero, there is a unique recursive solution to $r^n = 1 - as$. The binomial coefficients in (14.1) give this solution by the formal binomial identity. Alternatively, their recurrence verifies the differential identity $(1 - as)r' = -ar/n$, and hence that $r^n/(1 - as)$ has derivative zero and constant term 1. The normalized branch of its n th power is therefore as stated. Finally $r_{a,mn}^m$ has constant term 1 and its n th power is $1 - as$, so uniqueness proves (14.2). $\square$

Proposition 14.2 (Single-root degree). *For $a \neq 0$ and $n \geq 2$, the polynomial $X^n - (1 - as)$ is irreducible over $\mathbb{C}(s)$ and, when u is an order unit, over $\mathcal{F}_G(\mathbb{C})$. Its splitting field in the Hahn field is obtained by adjoining $r_{a,n}$ and has cyclic Galois group of order n .*

Proof. At the discrete valuation associated with $s = a^{-1}$, the element $1 - as$ has valuation 1. Eisenstein's criterion proves irreducibility over $\mathbb{C}(s)$. Transfer by Corollary 12.6 proves irreducibility over $\mathcal{F}_G(\mathbb{C})$. All roots are $\zeta r_{a,n}$ with $\zeta \in \mu_n \subset \mathbb{C}$, so the splitting field is the indicated cyclic extension. $\square$

[merge] The case $a = -1$, $n = 2$ is the example after Corollary 3.4: $(1 + s)^{1/2} = r_{-1,2}$ has degree two over $\mathcal{F}_G(L)$ for every characteristic-zero L .

14.2 Independence of all marked linear factors

Theorem 14.3 (Finite multiradical independence). *Let $a_1, \dots, a_q$ be distinct nonzero complex numbers and $n \geq 2$. Then*

$$[\mathbb{C}(s)(r_{a_1,n}, \dots, r_{a_q,n}) : \mathbb{C}(s)] = n^q, \quad \text{Gal} \cong (\mathbb{Z}/n\mathbb{Z})^q.$$

If u is an order unit, both conclusions remain true with $\mathbb{C}(s)$ replaced by $\mathcal{F}_G(\mathbb{C})$.

Proof. We prove the degree by adjoining the radicals in order. Suppose the first $j - 1$ have been adjoined and set $b = a_j^{-1}$, $w = s - b$. For every $i < j$, $1 - a_i(b + w)$ has nonzero constant term because $a_i \neq a_j$. It has n distinct n th roots in $\mathbb{C}[[w]]$: choose an n th root of the constant and apply the normalized binomial construction to the remaining unit.

Consequently the splitting field of the first $j - 1$ polynomials admits a $\mathbb{C}(s)$ -embedding into $\mathbb{C}((w))$. Use this embedding to restrict the w -adic valuation to $\Lambda_{j-1} = \mathbb{C}(s)(r_{a_1,n}, \dots, r_{a_{j-1},n})$. Its value group is $\mathbb{Z}$: it is a subgroup of $\mathbb{Z}$ and already contains the value 1 of $s - b$. In this valuation,

$$v_w(1 - a_j s) = 1.$$

Thus $X^n - (1 - a_j s)$ is Eisenstein over the valuation ring of Λ_{j-1} , and its degree over that field is n . This proves the inductive degree n^q .

The compositum is the splitting field of the product of these polynomials. Its automorphisms act by $r_{a_j,n} \mapsto \zeta_n^{e_j} r_{a_j,n}$ and embed the Galois group in $(\mathbb{Z}/n\mathbb{Z})^q$. Equality of orders makes this an isomorphism. Finally the whole compositum lies in $\mathbb{C}((s))$, so Corollary 12.6 preserves the result over $\mathcal{F}_G(\mathbb{C})$. $\square$

[merge] The w -adic valuation is written v_w to keep it apart from the Hahn valuation v ; source 02 writes v .

Remark 14.4 (A divisor interpretation). The same independence is visible in the valuations $\text{ord}_{s=a_j^{-1}}$. If $\prod_j (1 - a_j s)^{e_j}$ is an n th power in $\mathbb{C}(s)$, each e_j is divisible by n . This is the usual Kummer-theoretic description [13, Chapter 5, Kummer theory]. The proof above spells out the successive degree argument, including composite values of n .

14.3 The infinite tower

For $T \subseteq \mathbb{C}^\times$, let

$$E_T = \mathcal{F}_G(\mathbb{C})(r_{a,n} : a \in T, n \geq 1). \quad (14.3)$$

Choose the compatible roots of unity $\zeta_n = \exp(2\pi i/n)$ among the ordinary complex constants.

Theorem 14.5 (An explicit profinite quotient). *For an order-unit group G ,*

$$\text{Gal}(E_T/\mathcal{F}_G(\mathbb{C})) \cong \prod_{a \in T} \widehat{\mathbb{Z}}. \quad (14.4)$$

For infinite T ,

$$[E_T : \mathcal{F}_G(\mathbb{C})] = \max(|T|, \aleph_0), \quad |\text{Gal}(E_T/\mathcal{F}_G(\mathbb{C}))| = 2^{\max(|T|, \aleph_0)}.$$

In particular, for $T = \mathbb{C}^\times$ the extension has field degree $\mathfrak{c}$ and its Galois group has cardinality $2^{\mathfrak{c}}$.

Proof. An automorphism is determined by residues $e_{a,n} \in \mathbb{Z}/n\mathbb{Z}$ through

$$r_{a,n} \mapsto \zeta_n^{e_{a,n}} r_{a,n}.$$

The compatibility (14.2) makes these residues compatible when one integer divides another. Conversely, any compatible family acts on every finite subextension: choose a common multiple of its finitely many root orders and use Theorem 14.3. These actions agree on overlaps, so give an automorphism of E_T . The finite-level restriction maps give exactly the topology in (14.4).

There are at most $\max(|T|, \aleph_0)$ finite radical data, and each produces a finite-dimensional extension, giving that upper bound for the field degree. When T is infinite, the elements $r_{a,2}$, $a \in T$, are linearly independent over the base: every finite subset is part of the standard basis of a multiquadratic extension. This gives the lower bound $|T|$. If necessary, unbounded degrees in a single radical tower give the additional lower bound $\aleph_0$. Finally each $\widehat{\mathbb{Z}}$ has cardinality $\mathfrak{c}$, so the displayed group cardinality follows from cardinal exponentiation. $\square$

The comparison of infinite field degree with group cardinality is intentional. For $T = \mathbb{C}^\times$, the answer is $\mathfrak{c}$ for the former and $2^{\mathfrak{c}}$ for the latter. They must not be conflated.

14.4 Closedness without an order unit

The positive closedness assertion here is prior material of this report (Theorem 7.1 and Corollary 7.2). Source 02 includes its proof to make the opposite side of the Galois dichotomy transparent.

Remark 14.6 (Source 02's Proposition 7.1, printed once as Theorem 7.1). Source 02 states: *if G has no order unit, then*

$$\mathcal{B}_G(K) = \mathcal{F}_G(K) = \bigcup_{g>0} K((t^{C(g)})),$$

so $\mathcal{B}_G(K)$ is a field; conversely, if G has an order unit, $\mathcal{B}_G(K)$ is not a field. Here $C(g)$ is the principal convex subgroup of Section 7, divisible when G is. This is Theorem 7.1, and source 02's proof is that proof: a bounded well-ordered support lies in some $[-g, g]$, hence in $C(g)$; for g not an order unit some $h > 0$ exceeds every ng , so every series supported in $C(g)$ has support bounded above in G , whether or not that support is bounded inside $C(g)$; the principal convex subgroups form a directed family; and with an order unit u the inverse $\sum_{n \geq 0} t^{nu}$ of $1 - t^u$ has unbounded support. [merge] The union over the principal convex subgroups $C(g)$ equals the union over all proper convex subgroups in Theorem 7.1(2): every $C(g)$ with g not an order unit is proper, and a proper convex C lies in $C(g)$ for any positive $g \notin C$.

Theorem 14.7 (Exact closedness boundary). *For a nonzero divisible ordered abelian group G ,*

$$\mathcal{F}_G(\mathbb{C}) \text{ is algebraically closed} \iff G \text{ has no order unit,}$$

and

$$\mathcal{F}_G(\mathbb{R}) \text{ is real closed} \iff G \text{ has no order unit.}$$

Proof. Without an order unit, Remark 14.6 expresses the fraction field as a directed union of full Hahn fields over divisible convex subgroups. Each is algebraically closed for $K = \mathbb{C}$ and real closed for $K = \mathbb{R}$, by the classical Hahn-field theorem. A polynomial has only finitely many coefficients, all in one of these fields. This proves algebraic closedness in the complex case. In the real case, every positive element gets a square root and every odd-degree polynomial gets a root in one of the real-closed subfields, proving real closedness of the union.

With an order unit u , Proposition 14.2 for $a = 1$, $n = 2$ proves that $1 - t^u$ has no square root in $\mathcal{F}_G(\mathbb{C})$. It therefore has none in the real subfield either. But $1 - t^u$ is positive in that ordered field. This rules out both asserted forms of closedness. $\square$

[merge] The forward implications are Corollary 7.2, which allows any real closed K , and any algebraically closed K of characteristic zero. The reverse implications hold at that generality too: $X^2 - (1 - s)$ is irreducible over $K(s)$ for every field K of characteristic zero, since $1 - s$ has odd degree, so it stays irreducible over $\mathcal{F}_G(K)$ by Corollary 3.3(4) with $H = \mathbb{Z}u$ and $L = K$.

Example 14.8 (Countable cofinality is not the boundary). Let $G = \bigoplus_{n \geq 0} \mathbb{Q}e_n$, ordered by the largest index with nonzero coefficient, with all $e_n > 0$. Then $e_{n+1} > me_n$ for every positive integer m . The group has countable cofinality but no order unit. Hence $\mathcal{F}_G(\mathbb{C})$ is algebraically closed.

It is nevertheless smaller than $\mathbb{C}((t^G))$. The series $\sum_{n \geq 0} t^{e_n}$ is a legitimate Hahn series with an upper unbounded support. It does not belong to $\mathcal{B}_G(\mathbb{C})$, which in this case is already $\mathcal{F}_G(\mathbb{C})$. Thus closedness of the fraction field need not mean equality with the full Hahn field.

[merge] With $e_n = \omega^n$ this group is $G_{\aleph_0}$ of (22), and the example is Theorem 8.2(1) at $\kappa = \aleph_0$ together with the remark after Lemma 2.4 (countable cofinality does not give an order unit). The series $\sum_n t^{e_n}$ is even transcendental over $\mathcal{F}_G(\mathbb{C})$, by Corollary 4.6, and Theorem 8.2(2) gives $2^{\aleph_0}$ independent elements.

15 A nonsolvable family, real forms and cardinality

15.1 An explicit nonsolvable family

The geometric existence theorem can be complemented by one uniform polynomial family. Its classical monodromy calculation is elementary; it is included to make the surcomplex realization concrete.

Theorem 15.1 (Symmetric groups at a monomial scale). *Let $m \geq 2$, let u be an order unit, and put $s = t^u$. Then*

$$X^m - X - s$$

is irreducible over $\mathcal{F}_G(\mathbb{C})$, and its splitting field inside $\mathbb{C}((t^G))$ has Galois group S_m . All its roots belong to $\mathbb{C}[[s]]$. In particular $m = 5$ gives an explicit nonsolvable Galois extension.

Proof. First work over $\mathbb{C}(s)$. The polynomial map $\varphi : \mathbf{P}_x^1 \rightarrow \mathbf{P}_s^1$, $\varphi(x) = x^m - x$, has degree m . The corresponding extension of function fields has degree m , so $X^m - X - s$ is irreducible over $\mathbb{C}(s)$.

Its finite critical points satisfy $mc^{m-1} = 1$. They are nonzero and distinct, and $\varphi''(c) = m(m-1)c^{m-2} \neq 0$. Their critical values are

$$\varphi(c) = -\frac{m-1}{m}c,$$

so are also distinct. A small loop about each finite critical value therefore exchanges exactly two sheets. The finite local monodromies are transpositions. They generate the full monodromy group because a loop about infinity is the inverse of their product. The cover is connected, so the group they generate is transitive.

A transitive subgroup of S_m generated by transpositions is S_m : form the graph whose edges are the transpositions. Its orbits are the connected components, so it is connected; transpositions on the edges of a connected graph generate all permutations. Thus the splitting field over $\mathbb{C}(s)$ has group S_m .

At $s = 0$, the roots of $X^m - X$ are 0 and the $(m-1)$ st roots of unity, all distinct. At 0 the derivative is -1 , and at the other roots it is $m-1$. Formal coefficient recursion therefore lifts each root uniquely to $\mathbb{C}[[s]]$. The whole splitting field embeds into $\mathbb{C}((s))$. Apply Corollary 12.6 to preserve the polynomial's minimal degree and splitting-field Galois group over $\mathcal{F}_G(\mathbb{C})$. $\square$

Remark 15.2 (Small roots and large algebraic complexity). The root $x_0(s)$ with constant term 0 is determined by

$$x_0 = x_0^m - s.$$

For $m = 5$ it begins $-s - s^5 - 5s^9 - \dots$. It is a perfectly explicit surcomplex infinitesimal at $s = \omega^{-1}$. Its degree over the real-exponent complex omnific fraction field $\mathcal{F}_\mathbb{C} = \mathcal{F}_\mathbb{R}(\mathbb{C})$ is 5, and the normal closure of its field has group S_5 . Being represented by a convergent local complex series, or by a legitimate Hahn sum, does not make that series a bounded-support fraction.

15.2 Real multiquadratic extensions

Proposition 15.3. *Let u be an order unit and $T \subseteq \mathbb{R}_{>0}$. The field*

$$\mathcal{F}_G(\mathbb{R})(r_{a,2} : a \in T) \subseteq \mathbb{R}((t^G))$$

is Galois over $\mathcal{F}_G(\mathbb{R})$ with group $\prod_{a \in T} \mathbb{Z}/2\mathbb{Z}$.

Proof. For a finite subset $\{a_1, \dots, a_q\}$, the degree over $\mathbb{R}(s)$ is at most 2^q . Its complexification has degree 2^q over $\mathbb{C}(s)$ by Theorem 14.3, so its real degree is also 2^q . Alternatively, any smaller real linear relation would be a complex one. The real form of Corollary 12.6 preserves this degree over $\mathcal{F}_G(\mathbb{R})$. Sign changes of the square roots give the full Galois group. Passing to the directed union gives the product. $\square$

Every normalized root $r_{a,2}$ is positive in the Hahn order, because its constant term is 1. A nonidentity automorphism of one of these multiquadratic extensions need not preserve that order. There is no contradiction: these are abstract field automorphisms of fields that are not real closed. [merge] The restriction $T \subseteq \mathbb{R}_{>0}$ is source 02's; the proof uses only that each a is real and nonzero.

15.3 A generalized-dihedral radical extension

For $T \subseteq \mathbb{R}_{>0}$, consider the field E_T of (14.3). Coefficientwise conjugation fixes every $r_{a,n}$ and restricts to the nonidentity element of $\text{Gal}(\mathcal{F}_G(\mathbb{C})/\mathcal{F}_G(\mathbb{R}))$.

Theorem 15.4 (Real radical symmetry). *For an order-unit group G and $T \subseteq \mathbb{R}_{>0}$,*

$$\text{Gal}(E_T/\mathcal{F}_G(\mathbb{R})) \cong \left(\prod_{a \in T} \widehat{\mathbb{Z}} \right) \rtimes C_2,$$

where the nontrivial element of C_2 acts by inversion on every $\widehat{\mathbb{Z}}$ coordinate.

Proof. The field E_T is the splitting field, over $\mathcal{F}_G(\mathbb{R})$, of the family consisting of $X^2 + 1$ and all $X^n - (1 - at^u)$. Thus it is Galois. By Proposition 12.3 and Theorem 14.5, the kernel of restriction to $\mathcal{F}_G(\mathbb{C})$ is $\prod_{a \in T} \widehat{\mathbb{Z}}$. Conjugation supplies an order-two section of the quotient C_2 .

If $\sigma(r_{a,n}) = \zeta_n^{e_{a,n}} r_{a,n}$, conjugating this automorphism by coefficientwise conjugation changes ζ_n to ζ_n^{-1} and leaves $r_{a,n}$ fixed. The resulting coordinate is $-e_{a,n}$, proving the claimed semidirect action. $\square$

15.4 An exact cardinality calculation

Theorem 15.5 (Size of the absolute Galois group in real rank). *Let G be any nonzero additive subgroup of $\mathbb{R}$. Then*

$$|\mathcal{F}_G(\mathbb{C})| = |\mathcal{F}_G(\mathbb{R})| = \mathfrak{c}, \quad |\mathfrak{G}_{\mathcal{F}_G(\mathbb{C})}| = |\mathfrak{G}_{\mathcal{F}_G(\mathbb{R})}| = 2^{\mathfrak{c}}.$$

Each fraction field has exactly $\mathfrak{c}$ distinct quadratic extensions inside an algebraic closure. Divisibility of G is not necessary for this statement.

Proof. Every well-ordered subset of $\mathbb{R}$ is countable. One proof assigns to each non-last element of such a set a rational number strictly between it and its successor; these disjoint intervals give an injection into $\mathbb{Q}$. There is at most one last element. Therefore the Hahn supports in G are countable. The number of countable supports and coefficient assignments is at most $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$. Ordinary real or complex constants already give the reverse bound for the fraction fields.

Every positive $u \in G$ is an order unit. By Theorem 14.5, the complex absolute Galois group has a quotient of cardinality $2^{\mathfrak{c}}$. By Proposition 15.3, the real absolute Galois group has such a quotient as well. An algebraic closure of an infinite field of cardinality $\mathfrak{c}$ has cardinality $\mathfrak{c}$, and its automorphism group has cardinality at most $\mathfrak{c}^{\mathfrak{c}} = 2^{\mathfrak{c}}$. This proves equality.

The distinct marked square roots in Theorem 14.3 and Proposition 15.3 give $\mathfrak{c}$ distinct quadratic extensions: equality of two would contradict their degree-four compositum. There are at most $\mathfrak{c}$ quadratic polynomials over either base field, which proves the upper bound. $\square$

[merge] The count $|K((t^G))| \leq \mathfrak{c}$ is the one in the proof of Corollary 5.9.

No conclusion about the least number of topological generators follows from this cardinality calculation. Cardinality, topological weight, and profinite generator rank are different invariants.

16 One dominating scale splits all old finite extensions

16.1 Absorbing the entire old Hahn field

Let $H \subseteq G$ be an ordered subgroup inclusion. Say that $\theta \in G_{>0}$ *dominates* H if $\theta > h$ for every $h \in H$. In particular, $\theta > nh$ for every $h > 0$ in H and every ordinary positive integer n .

Theorem 16.1 (Uniform monomial denominator). *If θ dominates H , then for every coefficient field K ,*

$$K((t^H)) \subseteq \mathcal{B}_G(K).$$

More explicitly, for every $f \in K((t^H))$ and every unital subring $D \subseteq K$,

$$t^{-\theta} f \in K((t^{G^{<0}})), \quad f = \frac{t^{-\theta} f}{t^{-\theta}} \in \text{Frac } \mathcal{A}_G(D, K). \quad (16.1)$$

The same denominator $t^{-\theta}$ works for every old Hahn series.

Proof. The ordered inclusion preserves well-ordering of supports. Every old support is a subset of H and therefore bounded above by θ in G . Subtracting θ from its exponents puts the entire support in $G^{<0}$. Translation preserves well-ordering, giving both assertions in (16.1). $\square$

The upper bound here is external to the old group. This is precisely why an old unbounded support can become a numerator with only negative t -exponents. The support enlargement phenomenon itself is already illustrated in Section 8.4 of this report, and the displayed inclusion is (6) with $L = K$ [merge]. The next theorem makes its full Galois consequence explicit.

Theorem 16.2 (Simultaneous splitting of every old finite extension). *Assume H is divisible, and let $\theta \in G$ dominate it. Put $F = \mathcal{F}_H(\mathbb{C})$ and $F' = \mathcal{F}_G(\mathbb{C})$. Then F' contains an algebraic closure $\bar{F}$ of F . For every finite extension Λ/F there is an F' -algebra isomorphism*

$$\Lambda \otimes_F F' \cong (F')^{[\Lambda:F]}. \quad (16.2)$$

More generally every finite étale F -algebra becomes a finite product of copies of F' .

Proof. By the full Hahn-field closedness theorem, $\mathbb{C}((t^H))$ is algebraically closed. Let $\bar{F}$ be the subfield of elements of $\mathbb{C}((t^H))$ algebraic over F . This is an algebraic closure of F : the roots in $\mathbb{C}((t^H))$ of a polynomial with coefficients algebraic over F are themselves algebraic over F . By Theorem 16.1, $\mathbb{C}((t^H)) \subseteq F'$, so $\bar{F} \subseteq F'$.

Since the characteristic is zero, a finite extension Λ/F is separable and simple. Write $\Lambda = F(\alpha)$ with irreducible polynomial p of degree d . In $F'[X]$ the polynomial factors as

$$p(X) = \prod_{j=1}^d (X - \alpha_j)$$

with distinct roots $\alpha_j \in \bar{F} \subseteq F'$. The Chinese remainder theorem gives

$$\Lambda \otimes_F F' \cong F'[X]/(p) \cong \prod_{j=1}^d F',$$

proving (16.2). A finite étale algebra is a finite product of finite separable field extensions, so the final assertion follows. $\square$

For an embedded finite Galois extension $\Lambda/F \subseteq F'$, the isomorphism has the concrete form

$$x \otimes a \longmapsto (\sigma(x)a)_{\sigma \in \text{Gal}(\Lambda/F)}.$$

The different factors record the different old embeddings. Thus the symmetry is reflected in the splitting into factors; it is not erased from the abstract field Λ/F .

16.2 An exhaustive cofinality alternative

For an ordered subgroup inclusion there is no intermediate case between cofinality and the existence of a dominating element. Indeed, if $H \subseteq G$ is not cofinal, some $\theta \in G$ is strictly greater than every element of H . Since $0 \in H$, this θ is positive. [merge] This is the last assertion of Lemma 2.4.

Theorem 16.3 (All-or-nothing Galois transition). *Let $H \subseteq G$ be an inclusion of nonzero set-sized ordered abelian groups, with H divisible. Write $F = \mathcal{F}_H(\mathbb{C})$ and $F' = \mathcal{F}_G(\mathbb{C})$.*

1. *If H is cofinal in G , then for every finite extension Λ/F the tensor product $\Lambda \otimes_F F'$ is a field of degree $[\Lambda : F]$ over F' . If Λ/F is Galois, its Galois group and all intermediate-field degrees are preserved.*
2. *If H is not cofinal in G , then for every finite extension Λ/F ,*

$$\Lambda \otimes_F F' \cong (F')^{[\Lambda:F]}.$$

With compatible choices of algebraic closures, the restriction map $\mathfrak{G}_{F'} \rightarrow \mathfrak{G}_F$ is surjective in case (1) and trivial in case (2). When $\mathfrak{G}_F$ is trivial the two conclusions about this group homomorphism coincide; the criterion is a criterion on the exponent inclusion, not an assertion that these group properties are then disjoint.

Proof. Because H is divisible, $\mathbb{C}((t^H))$ is algebraically closed and contains an algebraic closure $\bar{F}$ of F . Every finite extension of F is isomorphic over F to a subfield of $\bar{F}$.

In the cofinal case, Theorem 3.2, with $K = L = \mathbb{C}$, shows that $\mathbb{C}((t^H))$ and F' are linearly disjoint over F . The same is true of each Λ . The tensor and Galois arguments of Corollary 12.6 apply with base F , proving (1). Restriction is surjective on every finite Galois quotient of $\mathfrak{G}_F$. Its image in $\mathfrak{G}_F$ is compact, hence closed, and these finite surjections make that image dense. It is therefore all of $\mathfrak{G}_F$.

In the noncofinal case, choose a dominating $\theta \in G$. Theorem 16.2 proves (2) and puts $\bar{F} \subseteq F'$. An automorphism fixing F' fixes $\bar{F}$ pointwise, so its image in $\mathfrak{G}_F$ is trivial. $\square$

Remark 16.4. The field-theoretic statements in the two cases are genuinely distinct when H has an order unit, since F then has nontrivial finite extensions. This provides a binary classification of how monomial exponent-group inclusions affect *all* finite Galois data at once. The cofinal support theorem is prior work; its combination with algebraic closedness of the old full Hahn field and noncofinal absorption is the Galois transition statement here.

Proposition 16.5 (The real counterpart). *If H is divisible and θ dominates it, then $\mathcal{F}_G(\mathbb{R})$ contains a real closure of $\mathcal{F}_H(\mathbb{R})$ compatible with its Hahn order.*

Proof. The field $\mathbb{R}((t^H))$ is real closed. Its relative algebraic subfield over $\mathcal{F}_H(\mathbb{R})$ is a real closure for the induced order: positive elements have square roots there, and odd-degree polynomials have roots there, because such roots in the ambient real closed field remain algebraic over the base. Apply Theorem 16.1. $\square$

The real statement does not say that every finite extension of the real base field splits after this base change. For example, $X^2 + 1$ still has no root in an ordered field. The complete splitting theorem is deliberately stated for complex coefficients.

16.3 The new group still has an order unit

Take the lexicographic extension

$$G = \mathbb{Q}\theta \oplus H,$$

where a nonzero θ -coordinate decides the sign before the H coordinate is considered. Then θ dominates H and is an order unit of G : every $q\theta + h$ lies between $-N\theta$ and $N\theta$ for an integer $N > |q| + 1$. If H is divisible, so is G .

Consequently Theorems 16.2 and 13.5 hold simultaneously. All old finite extensions split after this scalar extension, yet the new field has extensions realizing every finite group. When u is an order unit of H , the old missing root $\sqrt{1 - t^u}$ is now in the new fraction field, whereas the normalized root $\sqrt{1 - t^\theta}$ is not (Proposition 14.2 at the order unit θ). With Theorem 16.1, this proves Theorem 11.3.

Remark 16.6 (What is uniform). The scale θ depends on the old exponent group, not on an individual polynomial, a finite group, or a selected algebraic extension. A single θ absorbs the full old Hahn field and therefore all old algebraic extensions simultaneously. It does not absorb the full Hahn field over G itself.

17 Actual surreal examples and a countable rank tower

17.1 The normal-form embedding

Source 02 uses the Conway normal-form map (21), for an ordered additive subgroup $G \subseteq \mathbf{No}$, which identifies $\mathbb{R}((t^G))$ with a subfield of $\mathbf{No}$ and $\mathbb{C}((t^G))$ with a subfield of $\mathbf{No}[i]$ [15, 2, 1]; it is printed once, in Section 8.1. The image of $\mathcal{A}_G(\mathbb{Z}, \mathbb{R})$ consists of omnific integers: all growth exponents are positive except for an integer constant term. Likewise $\mathcal{A}_G(\mathbb{Z}[i], \mathbb{C})$ maps into $\mathbf{Oz}[i]$. Hence all the fraction-field distinctions of the Galois part are distinctions between actual surreal or surcomplex numbers. They are not constructions of a separate model with an unspecified embedding.

17.2 Real exponents: a root that is not an omnific fraction at that scale

Take $G = \mathbb{R}$ and $u = 1$, so $s = \omega^{-1}$, and consider the fields $\mathcal{F}_{\mathbb{R}} = \mathcal{F}_{\mathbb{R}}(\mathbb{R})$ and $\mathcal{F}_{\mathbb{C}} = \mathcal{F}_{\mathbb{R}}(\mathbb{C})$ of Section 8.2. The normalized square root has the Conway expansion

$$\sqrt{1 - \omega^{-1}} = 1 - \frac{1}{2}\omega^{-1} - \frac{1}{8}\omega^{-2} - \frac{1}{16}\omega^{-3} - \frac{5}{128}\omega^{-4} - \dots \quad (17.1)$$

It belongs to $\mathbb{R}((t^{\mathbb{R}}))$ but not to $\mathcal{F}_{\mathbb{R}}$, nor to $\mathcal{F}_{\mathbb{C}}$. Thus it cannot be represented as a fraction of two omnific integers all of whose growth exponents are real. The restriction on exponents is essential.

Now take $G' = \mathbb{Q}\omega + \mathbb{R}$. The exponent $\theta = \omega$ dominates all of $\mathbb{R}$. Formula (16.1) becomes

$$\sqrt{1 - \omega^{-1}} = \frac{\omega^\omega - \frac{1}{2}\omega^{\omega-1} - \frac{1}{8}\omega^{\omega-2} - \frac{1}{16}\omega^{\omega-3} - \dots}{\omega^\omega}. \quad (17.2)$$

The numerator is a legitimate omnific integer. Its exponents $\omega, \omega - 1, \omega - 2, \dots$ are decreasing and positive, and its coefficient at exponent 0 is 0. The denominator is also omnific. Both use exponents in G' .

However,

$$\sqrt{1 - \omega^{-\omega}} \notin \mathcal{F}_{G'}(\mathbb{R}),$$

and it does not belong to $\mathcal{F}_{G'}(\mathbb{C})$ either. The new order unit has recreated the same obstruction at the new scale.

[merge] The larger group is written G' because G is the generic exponent group. Section 8.4 uses $\mathbb{R}\omega + \mathbb{R}$ instead of $\mathbb{Q}\omega + \mathbb{R}$; ω dominates $\mathbb{R}$ and is an order unit there too, so the same two statements hold in that group.

17.3 An explicit direct limit of Galois-rich fields

Define ordered additive subgroups of **No** by

$$\Omega_0 = \mathbb{R}, \quad \Omega_n = \mathbb{R} + \mathbb{Q}\omega + \mathbb{Q}\omega^2 + \dots + \mathbb{Q}\omega^n \quad (n \geq 1), \quad \Omega_\infty = \bigcup_{n \geq 0} \Omega_n.$$

All sums here are finite within an element of a group.

Theorem 17.1 (A countable tower inside the surreal field). *For $K = \mathbb{R}$ or $\mathbb{C}$,*

$$\mathcal{F}_{\Omega_n}(K) \subsetneq K((t^{\Omega_n})) \subseteq \mathcal{F}_{\Omega_{n+1}}(K), \quad (17.3)$$

and

$$\bigcup_{n \geq 0} \mathcal{F}_{\Omega_n}(K) = \bigcup_{n \geq 0} K((t^{\Omega_n})) = \mathcal{F}_{\Omega_\infty}(K). \quad (17.4)$$

Every $\mathcal{F}_{\Omega_n}(\mathbb{C})$ realizes every finite group as a Galois group, and every finite extension of $\mathcal{F}_{\Omega_n}(\mathbb{C})$ splits over $\mathcal{F}_{\Omega_{n+1}}(\mathbb{C})$. The union in (17.4) is algebraically closed for $K = \mathbb{C}$ and real closed for $K = \mathbb{R}$. It is properly contained in $K((t^{\Omega_\infty}))$.

Proof. The group Ω_n is divisible and has order unit ω^n (with $\omega^0 = 1$). The exponent ω^{n+1} dominates all of Ω_n . The first inclusion in (17.3) is strict because the normalized root of $1 - t^{\omega^n}$ belongs to $K((t^{\Omega_n}))$ but not to $\mathcal{F}_{\Omega_n}(K)$. The second inclusion follows from Theorem 16.1. It already gives equality of the first two unions in (17.4).

The group Ω_∞ has no order unit. Any support bounded above and below in it is contained in some Ω_n : choose n large enough to contain its two bounds. An exponent with a nonzero coefficient at a higher power of ω would exceed one of those bounds in absolute order. Conversely, the whole group Ω_n is bounded above by ω^{n+1} in Ω_∞ . Hence

$$\mathcal{B}_{\Omega_\infty}(K) = \bigcup_n K((t^{\Omega_n})).$$

By Remark 14.6 this bounded ring is its own fraction field, proving (17.4). Closedness follows either from Theorem 14.7 or from the directed union of the closed fields $K((t^{\Omega_n}))$. Finite-group realization and splitting between successive stages follow from Theorems 13.5 and 16.2.

Finally the Hahn series

$$\sum_{n \geq 0} t^{\omega^n}$$

has well-ordered support unbounded in Ω_∞ . It belongs to $K((t^{\Omega_\infty}))$ but not to its bounded ring, which is the union in (17.4). $\square$

Under (21), the last witness is the actual surreal infinitesimal $\sum_{n \geq 0} \omega^{-\omega^n}$. Thus a proper algebraically closed subfield of $\mathbf{No}[i]$ can be the union of a countable increasing sequence of fields, each of which has the full finite-group realization property. The transition maps, rather than the abstract groups at the individual stages, are decisive.

[merge] The tower plays out, stage by stage, the enlargement of Section 8.4. Its witness is transcendental, not only outside the union: with $a_n = \omega^n$ the gap hypothesis of Corollary 4.6 holds in Ω_∞ (every $w \in \Omega_\infty$ is below some $N\omega^m$), so $\sum_n t^{\omega^n}$ is transcendental over $\mathcal{F}_{\Omega_\infty}(K)$, and Theorem 5.3 gives $2^{\aleph_0}$ algebraically independent elements over this algebraically closed union.

17.4 Why the full omnific field has no contradiction

Every set of surreal exponents has a surreal upper bound. For example, a Conway cut with that set on its left gives a larger surreal number. Thus for the full proper-class exponent group $\mathbf{No}$, every individual normal-form support is upper bounded. The formal bounded-support construction then gives the whole surreal or surcomplex field, with set-sized supports understood throughout. [merge] Section 8.4 makes the same observation; it is printed once there, with this argument added here.

The classical identity $\text{Frac } \mathbf{Oz} = \mathbf{No}$ is recovered by the same monomial-shift argument; it is not a new theorem here [1, Proposition 2.4.5 and its discussion of Conway's Theorem 32]. The nontrivial Galois phenomena above live in the specified set-sized fraction subfields. It is never asserted that $\mathbf{No}[i]$ itself has a nontrivial finite algebraic extension.

18 Scope, verification and relations of the Galois part

18.1 The role of divisibility

The finite-slice lemma, local descent, finite-group realizations, radical degrees, and uniform-denominator absorption hold for a general nonzero ordered abelian group subject to the stated order-unit or new-scale assumption. Divisibility enters the closedness dichotomy and the universal splitting theorem because those arguments use algebraic or real closedness of a full Hahn field.

Without divisibility, absence of an order unit does not by itself make the full convex-subgroup Hahn fields algebraically closed. Similarly, absorption of $\mathbb{C}((t^H))$ need not absorb an algebraic closure of the old field if this Hahn field is not algebraically closed. It still absorbs every explicitly constructed extension contained in that Hahn field.

18.2 Two unrelated senses of ramification

The points $s = a^{-1}$ used in the radical proof are valuations of function fields of complex algebraic curves. The Hahn valuation at $s = t^u$ instead records the least exponent of a series. Every normalized radical in (14.1) has Hahn valuation 0 and constant term 1. Its degree is nevertheless detected by ramification at the remote curve point a^{-1} before descent. This is not a contradiction; the two valuations have different domains of geometric interpretation.

In particular, the prime ideals of $\mathcal{A}_G(\mathbb{Z}, \mathbb{R})$, its integral closure, and ramification in that ring have not been classified. Those are ring-theoretic questions and require more than a function-field Galois group.

18.3 Finite groups versus the absolute Galois group

The Galois part gives a large explicit supply of quotients of $\mathfrak{G}_{\mathcal{F}_G(\mathbb{C})}$, and preserves a complete local Galois envelope. This does not determine the entire absolute Galois group. Its cohomological dimension, projectivity, freeness, and all finite embedding problems are not answered by the present proofs. The exact cardinality calculation in Theorem 15.5 must not be read as an isomorphism classification.

18.4 Support size and set theory

The dichotomy does not use CH, GCH, a measurable cardinal, or another large-cardinal axiom. The continuum bound in Theorem 13.5 comes from choosing independent branch points on the ordinary complex line. The results are not assertions that large cardinals have no effect on other surreal structures. Nor do they classify birthday-bounded or cardinally restricted Hahn fields. In a support-restricted setting, the formation of a binomial series, a field inverse, or an algebraic closure must be checked against the restriction.

18.5 Novelty and verification

The old descent theorem already implies the preservation of finite algebraic degrees. Source 02 does not relabel that implication as an independent breakthrough. Its proposed additional result is the systematic Galois theory obtained from it, including arbitrary independent finite-group products, explicit infinite radical groups, and uniform rank-transition splitting. The elementary pieces make these claims checkable, but do not certify historical priority. Its ledger and audit (Sections 18.8 and 18.9) distinguish the source inputs from new deductions.

18.6 What the Galois part does not claim

Every limitation stated by source 02, in its text, its delivery README or its audit, is kept; each also stands at its place above.

1. No nontrivial finite algebraic extension of $\mathbf{No}[i]$ is asserted. The Galois phenomena concern named set-sized fraction subfields, and every Galois group and tensor product is formed for set-sized fields; no proper-class absolute Galois group is defined.
2. $\mathcal{F}_G(K)$ is not identified with the full Hahn field $K((t^G))$ (Remark 12.2); the omnific fraction bridge is an equality with the bounded-support fraction field only.
3. The local Galois envelope $\mathcal{E}_G$ is not identified with an algebraic closure of $\mathcal{F}_G(\mathbb{C})$, and its group is not identified with the absolute Galois group.
4. The absolute Galois group $\mathfrak{G}_{\mathcal{F}_G(\mathbb{C})}$ is not determined: its cohomological dimension, projectivity, freeness and finite embedding problems are not answered, and it is not claimed to be a free profinite group. Realizing every finite group, and having cardinality 2^c , do not determine it.
5. Cardinality is not generator rank: Theorem 15.5 says nothing about the least number of topological generators or the topological weight, and infinite field degree (c) and group cardinality (2^c) are different numbers.

6. Nothing is said about the inverse Galois problem over $\mathbb{Q}$.
7. The prime ideals, integral closure and arithmetic ramification of $\mathcal{A}_G(\mathbb{Z}, \mathbb{R})$ and $\mathcal{A}_G(\mathbb{Z}[i], \mathbb{C})$ are not classified. The curve valuations at $s = a^{-1}$ and the Hahn valuation are two unrelated senses of ramification and are kept separate.
8. Divisibility is used only for closedness and for universal splitting. Without it only the extensions already inside the old Hahn field are absorbed, and absence of an order unit does not make the convex-subgroup Hahn fields closed.
9. The real case does not split every finite extension: $X^2 + 1$ has no root in an ordered field, and $\mathcal{F}_G(\mathbb{R})$ absorbs a real closure, not an algebraic closure (Proposition 16.5).
10. Splitting after scalar extension is not collapse of the old extension: Λ/F does not become equal to its base, and its automorphisms do not cease to exist.
11. The binary alternative of Theorem 16.3 concerns monomial inclusions of exponent groups only; general field embeddings of these fraction fields, possibly with intermediate behaviour, are open (Question 19.10).
12. A single dominating scale does not absorb the full Hahn field over the enlarged group itself (Remark 16.6).
13. No CH, GCH, measurable or other large cardinal is used, and nothing is asserted about the effect of large cardinals on other surreal structures.
14. Birthday-bounded and cardinally restricted Hahn fields are not treated.
15. Nothing is claimed in positive characteristic; the complex covering argument is not available there unchanged.
16. A description of the relative algebraic closure of $\mathcal{F}_G(\mathbb{C})$ in $\mathbb{C}((t^G))$ by a single local coordinate is not justified (Question 19.4), and embedding problems are solved only for independent products (Question 19.3).
17. The finite checks are regression tests of identities, signs, normalizations and finite algebra only. They do not prove cofinal descent, the existence of covers with arbitrary finite groups, infinite linear disjointness, cardinalities of absolute Galois groups, or any proper-class assertion, and no numerical approximation establishes a degree or a Galois group.
18. No Lean or other proof-assistant certificate exists, no repository build was performed by the source, and there has been no independent refereeing.
19. Historical priority is not certified. The support descent, the closedness without an order unit, Hahn closedness, the Riemann existence theorem, Kummer theory, transposition monodromy, Conway normal form and $\text{Frac } \mathbf{Oz} = \mathbf{No}$ are credited, not claimed; several conclusions are short consequences of known mechanisms once the right fraction fields are identified.
20. No named published conjecture is claimed solved, and L’Innocente–Mantova’s GCD and factorization conjectures are neither assumed nor claimed.
21. Source 02’s repository comparison was targeted (selected guides and audits at its pin, read through a connector with some responses truncated), not an exhaustive reading or build; its incomplete code search is not evidence of absence.
22. Byte-for-byte reproducibility of source 02’s PDF was not promised.

23. The research questions of Section 19 are not claimed to be open in the published literature.

18.7 Computational supplement

Source 02 ships a deterministic program using exact rational arithmetic and SymPy; here it is `code/02-galois-rank-verify.py`. It checks truncated binomial-power identities, compatibility of normalized radical branches, finite coset projection identities, discriminants of $X^m - X - s$ for $2 \leq m \leq 9$, local implicit-series identities, Chinese-remainder idempotents, finite permutation generation, finite generalized-dihedral identities, and finite lexicographic instances of the rank-shift construction. The delivered run (Python 3.13.5, SymPy 1.14.0, seed 20260923) passed 8,950 assertions in 18 named groups; its record is `data/02-galois-rank-verification_results.json`, and `data/02-galois-rank-verification_console.txt` is a byte-identical copy of it.

These tests do not prove the cofinal descent theorem, the existence of covers with arbitrary finite groups, infinite linear disjointness, cardinalities of absolute Galois groups, or any proper-class assertion. They are regression tests for signs, normalizations, and finite algebra used in the written proofs. No numerical approximation is used to establish a degree or a Galois group.

[merge] When the Galois part was merged, the unchanged script was run on a copy under Python 3.14.4 and SymPy 1.14.0 with an explicit `-output` path: all 8,950 assertions passed, and its output agreed with the delivered record in every field except the recorded Python version. Without `-output` the script writes `data/verification_results.json` beside its own `code/` directory, a file not shipped in this layout, so it should be run on a copy or with an explicit scratch path. The delivered `code/02-galois-rank-build.py` looks for `galois_rank_dichotomy.tex`, which is not shipped, and writes build transcripts into `data/`; `data/02-galois-rank-build_audit.json` records the delivered 27-page PDF, not this report's.

A practical proof-assistant implementation can be divided into four interfaces. First, prove that a bounded support has finite intersection with each coset of an order-unit cyclic subgroup. Second, show that coset projection intertwines multiplication by a series supported on that subgroup. Third, translate the resulting independence statement into injectivity of a field tensor product. Fourth, apply finite Galois base change. This organization does not require a full formalization of all surreal numbers before the central descent proof can be checked.

For the rank-jump theorem, an independent implementation needs only an ordered-group inclusion, a strict upper bound for its image, translation of a well-ordered support, and a verified algebraic-closedness theorem for the old full Hahn field. The Chinese-remainder splitting can then reuse ordinary field theory. This proof path is short enough to make formalization a realistic next verification step, but no such certificate accompanies the source manuscript, and no statement of the Galois part has an implementation mapping in the collection's formalization ledger.

18.8 Dependency and contribution ledger

Source 02's ledger, with its result numbers replaced by this report's. [merge] The entry for Corollary 12.6 is corrected: source 02 calls it "already implicit in prior degree preservation", but the degree and Galois-group assertions are explicit in Corollary 3.3 and the paragraph after it.

Result here	Essential input	Contribution status
Proposition 12.1	Translation of Hahn supports	Elementary bridge; not claimed new.

Result here	Essential input	Contribution status
Corollary 3.4 (source 02's Theorem 3.2; Remark 12.5) Corollary 12.6	Finite cyclic-coset slices Linear disjointness; finite Galois theory	Reproof of a special case of cofinal descent. Standard consequence; degree and Galois group explicit in Section 3 [merge], correspondence added.
Theorems 13.2, 13.3	Descent; Riemann existence; split local completion	Proposed application to the specified omnific fraction fields.
Theorem 13.5	Pairwise disjoint branch loci and joint descent	Proposed simultaneous realization package.
Theorems 14.3, 14.5	Eisenstein at marked curve points; formal binomial branches	Explicit degrees and profinite quotient over the named fraction fields; classical Kummer mechanisms credited.
Theorem 14.7	Prior bounded-ring theorem; Hahn closedness; single radical obstruction	Exact dichotomy, with the positive half already in this report.
Theorem 15.1	Classical transposition monodromy; descent	Explicit application, not a new S_m calculation over $\mathbb{C}(s)$.
Theorems 15.4, 15.5	Explicit towers; conjugation; cardinality bounds	Proposed real and cardinal refinements.
Theorem 16.1	A dominating exponent	Support phenomenon already present in this report; uniform formula restated.
Theorems 16.2, 16.3, 17.1	Cofinal descent; absorption; Hahn closedness; finite separability	Proposed exhaustive Galois transition and explicit rank-tower synthesis.

The ledger separates mathematical dependence from priority. A statement can be useful and have a complete proof while still being a short application of known results. Conversely, a claim of novelty would need a broader literature comparison than a successful proof alone provides.

18.9 Audit of possible failure points

Upper bounds must be internal to the named group. At an order unit, the support $\{nu : n \geq 0\}$ is unbounded in G , but becomes bounded after adjoining a dominating θ . Every use of boundedness specifies the ambient group.

Coset projection uses both ends of a support. A well-ordered support has a least exponent. Upper boundedness supplies the other end. Both are used to make a cyclic-coset slice finite. The result is false for arbitrary sets without a lower bound.

Degrees require linear disjointness. Finitely many denominators are cleared before a linear relation is projected. No attempt is made to project a quotient term by term. The full finite compositum is transported in the independent-family arguments.

The whole local splitting field is embedded. The finite cover is unramified over 0, and the residue field is $\mathbb{C}$. This makes its completion the base Laurent series field. Embedding a single ramified branch

would not suffice for the Galois-group statement.

The branch sets include infinity in their accounting. For the arbitrary-product construction infinity is unramified in every cover. The Kummer construction, which may have a common branch point at infinity, instead uses its separate marked-point Eisenstein proof. These two independence arguments are not interchangeable without checking their hypotheses.

No independence from coefficient transcendence is used. The radical coefficients can all be algebraic for rational a . Their algebraic obstruction comes from the support coordinate and function-field degree, not from choosing transcendental coefficients.

Divisibility is used exactly at closedness. The new group $\mathbb{Q}\theta \oplus H$ is divisible when H is. Full Hahn closedness then gives the old algebraic closure that is absorbed. Without this hypothesis only the extensions already present in the old Hahn field are automatically absorbed.

Infinite groups have their Krull topology. All displayed infinite product groups are topological isomorphisms, obtained from explicitly compatible finite quotient maps. Field degree and group cardinality are recorded separately.

The rank tower remains set-sized. Each Ω_n and Ω_∞ is a set, so ordinary field and Galois theory applies. The closing discussion of all of **No** is not used to define an unrestricted proper-class absolute Galois group.

Formal verification is not inferred from a repository. Although the repository contains Lean formalizations, source 02 was not checked by Lean and no repository build was performed in its preparation. Finite exact calculations are not promoted to proofs of infinite statements.

18.10 Relation to neighbouring reports

[merge] This subsection compares statements only; no theorem of another report is used in a proof of the Galois part.

- *Tail-spans and differential transcendence* [5]. Its `tail:thm:closure` describes the valuation closure of the relative algebraic closure of a *full* Hahn field $M((t^\Gamma))$ inside $\overline{M}((t^\Gamma))$, and its noncofinal example $\mathbb{Q} \hookrightarrow \mathbb{Q}^2$ loses algebraic approximability. Here the base is the bounded fraction field and the noncofinal enlargement of Theorem 16.3 splits every finite extension. Both illustrate that a noncofinal embedding changes algebraic answers for the same series; the objects differ, and Question 19.4 asks for the relative algebraic closure over the bounded base, which that report does not treat.
- *Entire functions at arbitrary rank* [6]. Its `ent:thm:main-extension(b)`: a nonpolynomial entire series over $\mathbb{C}((t^\Gamma))$ remains entire over a larger group exactly when the old group is cofinal in it. Theorem 16.3: a finite extension of $\mathcal{F}_H(\mathbb{C})$ remains a field of the same degree over $\mathcal{F}_G(\mathbb{C})$ exactly when H is cofinal in G , and splits completely otherwise. The same cofinal/noncofinal dichotomy, for different objects, with independent proofs.
- *Independent surreal copies* [16]. Its `isc:prop:boundedfrac` is Proposition 12.1 for $D = \mathbb{Z}, \mathbb{Z}[i]$ (written after source 02's pin); its coset slices are the projections used in Remarks 12.5 and 12.8.
- *Cardinally restricted fields*. Question 19.9 concerns fields of Hahn series with fewer than κ terms. Such fields are studied in the reports *The First- κ Coefficients* [11] and *Critical-Point Defects in Surreal Arithmetic* [12], whose short-support field $\mathcal{S}_{<\kappa}$ is of this kind. Nothing from them is used here, and no answer to the question is claimed from them.

19 Research questions from the Galois part

The following are proposed research problems motivated by the proved results of source 02. They are not all asserted to be open in the wider literature; settling their exact prior status is part of the first task in pursuing them. [merge] Source 02 prints them as the twelve subsections of its Section 13; their text is kept, in this report's notation.

Question 19.1 (The complete absolute Galois group). Determine $\mathfrak{G}_{\mathcal{F}_G(\mathbb{C})}$ for a divisible order-unit group G , beginning with $G = \mathbb{Q}$ and $G = \mathbb{R}$. Is its isomorphism type controlled by the top Archimedean quotient, by the lower convex subgroups, or by finer support structure? A first useful invariant would be its cohomological dimension. Realizing every finite group and having cardinality 2^c do not answer this question.

Question 19.2 (The kernel outside the local envelope). Study the restriction epimorphism

$$\mathfrak{G}_{\mathcal{F}_G(\mathbb{C})} \longrightarrow \text{Gal}(\mathcal{E}_0/\mathbb{C}(s))$$

from Theorem 13.3. Describe its kernel and compare the images coming from different order units. Existing results about fundamental groups of complex curves may identify the target more explicitly, but they do not automatically identify the kernel.

Question 19.3 (Finite embedding problems). Given a finite quotient of $\mathfrak{G}_{\mathcal{F}_G(\mathbb{C})}$ and an epimorphism onto that quotient from another finite group, determine when a compatible proper solution exists. The construction by disjoint branch loci solves independent product problems, not arbitrary prescribed compatibility problems. A useful intermediate target is an embedding problem entirely contained in the local envelope.

Question 19.4 (The relative algebraic closure inside the Hahn field). Give an intrinsic support description of the elements of $\mathbb{C}((t^G))$ algebraic over $\mathcal{F}_G(\mathbb{C})$. The local algebraic series in t^u provide many examples, but the coefficients of a polynomial over the fraction field may involve infinitely many cosets. A description using only a single local coordinate is not justified by source 02.

Question 19.5 (Minimal enlargement for a prescribed root). For $f \in K((t^G))$, classify the smallest ordered group extensions in which f becomes a fraction of canonical omnific integers. A dominating element always suffices, but may be much more than is needed for a particular support. An exact criterion for finite families of algebraic branches would sharpen the uniform denominator construction.

Question 19.6 (Integral closure and arithmetic ramification). Let $E/\mathcal{F}_G(\mathbb{C})$ be one of the explicit cyclic or symmetric extensions. Determine the integral closure of $\mathcal{A}_G(\mathbb{Z}[i], \mathbb{C})$ in E , and identify the primes at which this ring extension ramifies. The curve valuations that prove independence are an input, not an already established classification of omnific prime ideals.

Question 19.7 (Real realization with prescribed orderings). Beyond the multiquadratic and generalized-dihedral families, which finite Galois groups can be realized inside $\mathbb{R}((t^G))$ over $\mathcal{F}_G(\mathbb{R})$? How many orderings extend the base Hahn order? Real covers with carefully chosen split real fibers should give an approach, but complex realization alone does not prove a real theorem.

Question 19.8 (Nondivisible groups and positive characteristic). Separate the contribution of missing divisibility from the contribution of an order unit. In positive characteristic, replace the radical towers by suitable Artin–Schreier or Witt towers and analyze wild ramification. The complex covering argument is not available unchanged, and a naive characteristic-zero transcription would be invalid.

Question 19.9 (Cardinally restricted supports). Repeat the constructions in fields of Hahn series with fewer than κ terms. Which steps are preserved by regular uncountable κ , and what happens at singular cardinals? It is necessary to check both algebraic closure operations and the supports of all coefficients appearing in a finite cover. Counting the displayed radicals alone is not sufficient.

Question 19.10 (Functorial rank transitions). Develop a category of exponent groups together with the induced fraction fields and finite étale algebras. A dominating extension sends every finite étale algebra from the old divisible complex stage to a split algebra. Theorem 16.3 settles the binary behavior for ordered exponent-group inclusions. Which more general embeddings of these fraction fields can exhibit intermediate behavior, splitting some finite extensions but preserving others? Distinguishing monomial embeddings from general field embeddings is essential here.

Question 19.11 (Effective computation and certificates). Build algorithms that take a polynomial with a split simple fiber and produce certified local series, degree information, and the resulting Hahn-field embedding. Explicit monodromy or group certificates would be needed for a general finite cover. Such an algorithm should not claim to decide equality of arbitrary surreal presentations.

Question 19.12 (Formalization of the support-to-Galois bridge). Formalize the finite-slice lemma and the tensor injectivity proof before attempting the geometric existence theorem. This gives a useful theorem in any library with Hahn series, ordered groups, and field extensions. A second stage can formalize the explicit radical families and rank absorption entirely algebraically. The Riemann-existence input can then be isolated behind an exact theorem interface rather than silently assumed by the code.

A Normalized GCD descent and a published question

This appendix is logically independent of the transcendence theorems. It provides a full proof of a conditional reduction rather than importing a recent claimed result as an established premise.

A.1 The input and the conclusion

For a characteristic-zero field K , let $\bar{K}$ be an algebraic closure. Consider the hypothesis

$$\mathcal{B}_{\mathbb{R}}(\bar{K}) \text{ is a GCD domain.} \quad (\text{GCD}_{\mathbb{R}, \bar{K}})$$

A GCD domain is an integral domain in which every pair has a greatest common divisor, understood up to units. The existence of a GCD for a finite nonempty family follows by induction. By Proposition 7.4, the units of the rank-one bounded-support ring are exactly the nonzero monomials.

For a nonzero bounded series define

$$N(f) = \text{lc}(f)^{-1} t^{-v(f)} f. \quad (24)$$

It has least exponent zero and coefficient 1 there. Two nonzero bounded series are associates precisely when their normalizations agree.

Theorem A.1 (Normalized GCD descent). *Assume $(\text{GCD}_{\mathbb{R}, \bar{K}})$. For every nonzero ordered subgroup $H \subseteq \mathbb{R}$, $\mathcal{B}_H(K)$ is a GCD domain. More precisely, the normalized GCD in $\mathcal{B}_{\mathbb{R}}(\bar{K})$ of any finite family from $\mathcal{B}_H(K)$, not all zero, belongs to $\mathcal{B}_H(K)$ and is a GCD there. No divisibility or completeness assumption on H is required.*

A.2 Characters separate exponent cosets

Lemma A.2. *Let A be an abelian group and L an algebraically closed field of characteristic zero. For every $0 \neq a \in A$, there is a character $\chi : A \rightarrow L^\times$ with $\chi(a) \neq 1$.*

Proof. On the cyclic group generated by a , choose a primitive m th root of unity as the image of a if it has finite order $m > 1$, and choose 2 if it has infinite order. In either case the resulting character separates a from zero.

We explain why this character extends. The abelian group $L^\times$ is divisible, because every nonzero scalar has an n th root. Suppose a character has been defined on a subgroup $D \subseteq A$, and choose $x \notin D$. If no nonzero integer multiple of x lies in D , give x any nonzero image. Otherwise the integers n with $nx \in D$ form $m\mathbb{Z}$ for a least positive m . Choose $y \in L^\times$ with $y^m = \chi(mx)$, and set

$$\tilde{\chi}(d + nx) = \chi(d)y^n.$$

If $d + nx = d' + n'x$, then $(n - n')x = d' - d \in D$. In the nontrivial case $n - n' = km$, so $\chi(d') = \chi(d)\chi(mx)^k = \chi(d)y^{n-n'}$ and the assigned values agree. In the other case the representation is unique. The formula is multiplicative with respect to addition, so it is an extension. Order the extensions by domain inclusion. Their domains are subgroups of the set A , and the union along a chain is again a character extending the given one. Zorn's lemma gives a maximal extension; the preceding one-element construction forces its domain to be all of A . $\square$

Proof of Theorem A.1. Let d be the normalized ambient GCD of $f_1, \dots, f_m \in \mathcal{B}_H(K)$. For every character $\chi : \mathbb{R}/H \rightarrow \overline{K}^\times$, define

$$T_\chi \left(\sum_r a_r t^r \right) = \sum_r a_r \chi(r + H) t^r.$$

This is an automorphism of the Hahn field and of its bounded-support subring: it preserves support and its multiplicativity follows at each finite convolution coefficient. Its inverse uses χ^{-1} . It takes normalized series to normalized series, since the coefficient at exponent zero remains 1. It fixes every f_j . Consequently $T_\chi(d)$ is another normalized GCD of the same family. Uniqueness of normalization implies $T_\chi(d) = d$.

If an exponent $r \notin H$ occurred in d , Lemma A.2 would supply a character with $\chi(r + H) \neq 1$. Its nonzero coefficient would then change, contradicting the equality. Hence $\text{supp}(d) \subseteq H$.

Next every automorphism in $\text{Aut}(\overline{K}/K)$ acts coefficientwise and again preserves normalization and fixes the input family. Thus it fixes every coefficient of d . In characteristic zero the fixed field of all these automorphisms is K . Indeed, an algebraic element outside K has a separable minimal polynomial with another conjugate; an embedding taking it to that conjugate extends to an automorphism of the algebraic closure. Therefore every coefficient of d lies in K . Its support is bounded in $\mathbb{R}$ and lies in the nonzero subgroup H , which is cofinal in $\mathbb{R}$; it is bounded in H . Thus $d \in \mathcal{B}_H(K)$.

Each quotient f_j/d belongs to $K((t^H))$ because this is a field. It is bounded in the ambient ring by divisibility of the ambient GCD, so cofinality makes it bounded in H . Hence d divides every f_j in the base ring. If $e \in \mathcal{B}_H(K)$ is any common divisor, then $e \neq 0$ because at least one input is nonzero. Ambient universality gives $e \mid d$ there. The quotient d/e is again in $K((t^H))$ and ambient bounded, hence in $\mathcal{B}_H(K)$. This proves base-ring universality. The all-zero family can separately be assigned GCD zero. $\square$

A.3 Relation to the pre-Schreier question

L’Innocente and Mantova ask, immediately after their Example 9.1.3, whether $\mathcal{B}_H(K)$ is pre-Schreier for a divisible Archimedean group H [1]. Since a GCD domain is pre-Schreier, Theorem A.1 gives a positive answer *conditional on* $(\text{GCD}_{\mathbb{R}, \bar{K}})$. The latter hypothesis also follows if the one-sided ring $\bar{K}((t^{\mathbb{R}^{\leq 0}}))$ is a GCD domain, since Proposition 2.2 identifies its monomial localization with $\mathcal{B}_{\mathbb{R}}(\bar{K})$, and localization preserves the GCD property. These two general commutative-algebra implications are standard; the new argument in this appendix is the normalized character descent.

There is a further conditional consequence at arbitrary divisible rank. Suppose the real-exponent GCD input holds over the algebraic closure of *every* characteristic-zero coefficient field. If G has no order unit, its bounded-support ring is already a field. If it has an order unit, Proposition 7.3 writes its bounded ring as $\mathcal{B}_H(M)$ with $H \subseteq \mathbb{R}$ and $M = K((t^{\mathbb{C}}))$. Applying the descent theorem to this coefficient field proves that $\mathcal{B}_G(K)$ is GCD. The required ambient hypothesis is now about $\mathcal{B}_{\mathbb{R}}(\bar{M})$, where $M = K((t^{\mathbb{C}}))$; an input only for $\bar{K}$ would not suffice. Thus a universal rank-one input would settle bounded-support GCD arithmetic at every divisible rank.

Warning A.3 (Recent external claim, not a premise of the main results). The source audit records that Dan Abramov’s public 2026 project gaearon/conway-refinement advertised a Lean proof that the one-sided real-exponent Hahn ring is GCD [10]. Its standalone statement `HahnSeriesGCD.lean` has a characteristic-zero hypothesis and the relevant nonpositive-support definition. The project described its work as provisional at that audit. The source audit read the statement and README but did not build or verify the full proof dependency chain or independently validate its semantics. Accordingly this article retains $(\text{GCD}_{\mathbb{R}, \bar{K}})$ as an explicit hypothesis. It does not claim an independent resolution of Conway’s refinement conjecture or an unconditional resolution of the published bounded-support pre-Schreier question.

B Detailed audit and dependency ledger

B.1 Novelty candidates versus imported facts

Statement	Status in this draft	Dependency boundary
Cofinal coefficient-exponent descent	Proposed contribution; full proof in Theorem 3.2.	Coset extraction, a coefficient-linear functional, finite convolution.
Exact cofinality converse	Proposed companion; Sections 3 and 5.	Noncofinal supports become bounded; properness follows from independence.
$2^{\text{cf}(G)}$ independent family on optimal support	Principal proposed contribution; Theorem 5.3.	Full support-gap and coding proofs; no GCD input.
Earlier nonrationality/cofinality obstruction	Prior result, not claimed new.	L’Innocente–Mantova, Proposition 2.4.5.
Finite-pattern coding and integer-grid detection	Elementary background mechanisms, proved here.	Choice and finite polynomial root bounds.
Real-exponent degree $\mathfrak{c}$ and arbitrary-regular-cardinal surreal examples	Consequences of the principal theorem, with explicit constructions.	Independent cardinal upper bounds; standard normal form for actual surreals.

Statement	Status in this draft	Dependency boundary
Field criterion and top-rank unit formulation	Structural companion results; modest novelty claim.	Hahn fields; top-rank grouping is standard; rank-one upper-support theorem imported.
Hahn real/algebraic closedness and Conway normal form	Imported foundations, not new proofs.	Standard results explicitly identified in the text.
Normalized GCD descent	Proposed conditional reduction; full implication proof.	Existence of the ambient GCD remains a hypothesis.
Lean certification of this article	Not claimed.	Finite exact tests are not a proof-assistant check.
Comparison with other reports (Section 9)	Editorial addition at placement; no new theorem.	Statements of the cited reports, quoted by label; none is used in the proof of a theorem here.
Galois part, Sections 11–19 (batch 32)	Second source; its own contribution ledger is Section 18.8.	Descent of Section 3; Riemann existence; Hahn closedness.

B.2 Repository comparison actually performed

The repository’s root overview, documentation catalogue, current docs/new tree, and the tail-span report’s detailed README were read through the GitHub connector. The initial inspected revision was `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0`. The docs/new tree at that revision contained its README rather than an additional unreviewed article collection. Targeted repository searches for lacunary and bounded support returned no indexed matches; that negative search result alone is not proof of absence. The comparison relies principally on the catalogue and the closest report’s explicitly described base fields and conclusions.

The audit was not a line-by-line comparison against all 36 full manuscripts, all retired sources in Git history, or all Lean declarations. It therefore supports the formulation “not identified in the reviewed material,” not a guaranteed absence from every historical file. The delivered `research_audit.md` records this limitation alongside the literature queries and the exact mathematical strengthening over the closest published cofinality result.

B.3 Repository check made when the report was placed

The placement and assembly history is now checked against the retained commits `30dfb4f` and `ed88b8f`. Revision `048b72cf7cbfc8ab246e4f73788c10460cb3f6e0` exists; its catalogue lists 36 reports, and the later revision `6e34651fe640` is its successor, concerning Hahn–Hilbert restrictions and closed range. At the pin, a case-insensitive literal search in the tracked documentation, excluding docs/new, finds no lacunary. The phrase “bounded support” has two textual matches, in the analytic-geometry guide and article, describing the same unrelated “bounded support cuts” topic in a fibre of maximal ideals.

At assembly, lacunary does occur in this newly added report and its delivered audit, so a claim that it occurs nowhere in that tree would be false. The same search outside this report still finds none at that specific commit. Separately, the three-duals report’s `duals:ex:bounded-support` discusses a vector-valued Hahn series with support bounded above. That is a conceptual comparison, not a literal hit for the phrase “bounded support” at assembly. It concerns approximation in Hahn vector

spaces, not the ring (1) or its fraction field. These dated searches make no absence claim about the current or future collection; the mathematical comparisons are in Section 9.

B.4 Proof audit: the points most vulnerable to mistakes

The proofs depend on several distinctions that deserve an explicit final check. The base is the fraction field of bounded-support series, so every putative relation must first have its denominators cleared. At limit indices the earlier partial series are not finite; the proof uses bounded support, not finite support. A top homogeneous polynomial need not be nonzero on an arbitrary coefficient sequence, which is why the coding lemma realizes a complete positive-integer grid cofinally. The coding family has size 2^κ but uses only κ positions; finite patterns, not arbitrary infinite patterns, are what it realizes.

Coset projection is not a ring homomorphism; it is linear over the Hahn field on the subgroup. Likewise the coefficient functional is not a field homomorphism; it is linear over the smaller coefficient field. Cofinality is exactly what makes their output coefficients bounded in the smaller group. For units, upper-support multiplicativity is used only after passing to an Archimedean quotient. For concrete surreal examples, the monomial map is Conway's $x \mapsto \omega^x$, and the Hahn valuation sign is reversed in (21). Finally, a maximal-cardinality independent family is not claimed to be a transcendence basis.

B.5 What the deliverable establishes

The unconditional core is a collection of relative algebraic theorems with detailed proofs and explicit witnesses, including the sharp support threshold, coefficient–exponent descent, and arbitrarily large actual surreal examples. Their mathematical validity can be assessed directly from the given arguments and the named standard foundations. Their publication novelty requires further expert literature review. The optional GCD reduction has an additional stated hypothesis and should not be promoted into an unconditional open-problem solution by silently accepting an unverified external claim.

References

- [1] Sonia L’Innocente and Vincenzo Mantova, *A factorisation theory for generalised power series and omnific integers*, *Advances in Mathematics* **442** (2024), article 109513. doi:10.1016/j.aim.2024.109513. Version examined: arXiv:1710.07304v5, 22 January 2024. Relevant locations: Fact 2.1.1, Section 2.3, Fact 2.4.2, Proposition 2.4.5, Proposition 3.5.1, and the question after Example 9.1.3.
- [2] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. doi:10.1017/CBO9780511629143. Cited for standard normal-form foundations; bibliographic data also recorded in [1].
- [3] The Stacks Project Authors, *The Stacks Project*, Section 9.27, “Linearly disjoint extensions,” Tag 09IC. Accessed 22 September 2026.
- [4] Vladimir Reshetnikov, *Surreal*, repository documentation catalogue, docs/README.md, revision 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0. Versioned repository source. Accessed 22 September 2026.
- [5] *Tail-spans and differential transcendence in surreal and surcomplex Hahn fields*, AI-assisted research report in VladimirReshetnikov/Surreal,

- docs/surreal/tail-spans-and-differential-transcendence/. Detailed README examined at the same revision as [4]. [Versioned report overview](#). Not treated as a refereed source. Labels quoted in Section 9 were checked against its `article.tex` at assembly commit `ed88b8f`; the current comparison was revisited as stated in Section 9.
- [6] *Cofinality, Factorization and Scalar Extension for Entire Hahn Functions at Arbitrary Rank*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/surcomplex/entire-functions-at-arbitrary-rank/. Labels quoted in Section 9 were checked at assembly commit `ed88b8f` and revisited as stated there. Not treated as a refereed source.
 - [7] *Hidden Negative Directions in Surcomplex Linear Algebra*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/surcomplex/hidden-negative-hermitian-directions/, placed in commit `30dfb4f` and assembled in `ed88b8f`. Cited here by theorem title (“Independent prime-tail family”) and by the definition of its base field $\mathcal{P}_T$. Not treated as a refereed source.
 - [8] Notation guide of VladimirReshetnikov/Surreal, docs/NOTATION.md, section “Topologies and strong summation” and the table “Scalars, workspaces, and size”, at assembly commit `ed88b8f`.
 - [9] Implementation note docs/NORMAL_FORM_BRIDGE.md of VladimirReshetnikov/Surreal, on the Lean normal-form bridge, at assembly commit `ed88b8f`.
 - [10] Dan Abramov, *A Proof of Conway’s Refinement Conjecture*, public provisional project gaearon/conway-refinement, 2026. [Project README](#) and [standalone Hahn-series GCD statement](#). Accessed 22 September 2026. The proof build was not independently rerun in this audit; the stated GCD input is retained as a hypothesis.
 - [11] *The First- κ Coefficients*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/surcomplex/first-kappa-coefficients/. Cited by name in the Galois part (batch 32); no label or result of it is used. Not treated as a refereed source.
 - [12] *Critical-Point Defects in Surreal Arithmetic*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms/. Cited by name in the Galois part (batch 32) for its short-support field; no label or result of it is used. Not treated as a refereed source.
 - [13] James S. Milne, *Fields and Galois Theory*, version 5.10, September 2022, 144 pp. [Author’s course notes](#). Chapters 3, 5, 7, and 8 provide the finite, Kummer, infinite, and étale-algebra background used in the Galois part (source 02).
 - [14] James S. Milne, *Lectures on Étale Cohomology*, version 2.21, 22 March 2013, 202 pp. [Author’s course notes](#). Theorem 21.3 is the Riemann existence theorem; the account refers to SGA 1, Exposé XII. Cited by source 02.
 - [15] John H. Conway, *On Numbers and Games*, second edition, A K Peters, 2001 (first edition, 1976). Classical background reference, cited by source 02; not newly reviewed in full for it.
 - [16] *Independent Surreal Copies and Arithmetic Intersections*, AI-assisted research report in VladimirReshetnikov/Surreal, docs/surreal/independent-surreal-copies/, written in commit `781b19e` (batch 31). Labels quoted after Corollary 4.6 and in Section 9.5 were checked against its `article.tex` at that commit. Not treated as a refereed source.